

Polarization- and Mutual-Coupling-Aware Rotatable Antennas Enabled Wireless Communications: Modeling, Orientation Optimization, and Prototyping

Jianchuan Wei^{ID}, Simin Song^{ID}, Zhenyu Kang, Tengjiao Wang^{ID}, *Member, IEEE*, Huaqiang Gao^{ID}, *Member, IEEE*, Shanpu Shen^{ID}, *Senior Member, IEEE*, and Xiaoming Chen^{ID}, *Senior Member, IEEE*

Abstract—Rotatable antennas (RAs) expand the spatial degrees of freedom of multiple-input multiple-output (MIMO) systems beyond conventional fixed-panel arrays. However, existing evaluations often model rotation merely as a scalar gain adjustment and overlook three hardware-critical effects: 1) the phase carried by the antenna pattern, 2) rotation-induced changes in the polarization, and 3) orientation-dependent mutual coupling variations. In this context, we develop a unified modeling-evaluation-optimization framework that explicitly incorporates phase, polarization, and mutual coupling effects through a rotation-explicit complex vector element pattern (CVEP). Polarization coupling and mismatch are captured by embedding the rotated CVEPs into a 2×2 per-ray polarization structure, while mutual coupling variation is modeled via power-consistent interpolation of embedded radiation patterns under joint orientation states. Additionally, a compact RA base-station (RA-BS) element and a 6×3 RA-BS prototype supporting $\pm 60^\circ$ mechanical rotation with a high front-to-back ratio are designed, assembled, and measured. The simulation and measurement results validate that the proposed electromagnetic (EM)-consistent RA system model accurately characterizes rotation-induced variations in gain, polarization, and radiation patterns. Furthermore, an alternating feasible-direction optimization algorithm, tailored to the jointly programmable BS- and subarray-level rotations, is developed to maximize the downlink sum-rate in multi-user MIMO deployments. Our simulations demonstrate that the proposed rotation-aware RA-BS architecture significantly outperforms conventional gain-only and mutual-coupling-neglected baselines. These findings highlight the necessity of incorporating phase, polarization, and mutual coupling effects into RA modeling and optimization

to ensure robust performance under uniform user distributions and maximize achievable throughput in hotspot deployments.

Index Terms—Rotatable antenna, complex vector element pattern, polarization, mutual coupling, orientation optimization.

I. INTRODUCTION

NEXT-GENERATION wireless networks demand unprecedented capacity, flexibility, and spatial reconfigurability, motivating the exploration of new antenna architectures that offer additional spatial degrees of freedom (DoF) [1], [2].

Traditional multiple-input multiple-output (MIMO) systems are being advanced by emerging technologies such as fluid antennas [3], [4], [5] and movable antennas [6], [7], [8], which dynamically adjust their shape or position through reconfigurable liquid materials [9], conductive elements [10], or mechanical actuators [11]. These approaches illustrate the broader trend of six-dimensional movable antennas (6DMAs) [12], [13], [14], [15], [16], [17], where an antenna or array can be moved in three-dimensional (3D) space and/or rotated in 3D orientation to fully leverage channel spatial variations. However, fully positionable antennas entail high complexity and cost due to mechanical constraints. This has led to growing interest in rotatable antennas (RAs) [18], [19], [20], [21], [22], [23] as a more practical subset of 6DMA. As shown in Fig. 1, in an RA system, antennas remain fixed in place but can rotate about one or more axes, greatly simplifying hardware while still enabling radiation pattern reconfiguration.

Notably, RAs have been explored in various emerging scenarios. In integrated sensing and communications (ISAC), RAs enhance joint beamforming and sensing by optimizing array rotations at a single base station (BS) or across multi-BS cooperative ISAC networks, where rotation angles and transmit beams are designed to meet desired beam patterns under SINR/power constraints [20]. In physical-layer security, RA-based arrays adjust per-element boresights to improve the legitimate link while suppressing eavesdroppers, resulting in higher secrecy rates than fixed-antenna benchmarks [21]. Beyond far-field settings, rotatable/movable antenna platforms apply RA concepts to near-field ISAC by jointly

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Jianchuan Wei, Simin Song, Huaqiang Gao, and Xiaoming Chen are with the School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: xiaoming.chen@mail.xjtu.edu.cn).

Zhenyu Kang and Tengjiao Wang are with the Department of Wireless Network Research, Huawei Technologies Company Ltd., Shanghai 201206, China (e-mail: kangzhenyu@huawei.com; wangtengjiao6@huawei.com).

Shanpu Shen is with the State Key Laboratory of Internet of Things for Smart City and the Department of Electrical and Computer Engineering, University of Macau, Macau, China (e-mail: shanpushen@um.edu.mo).

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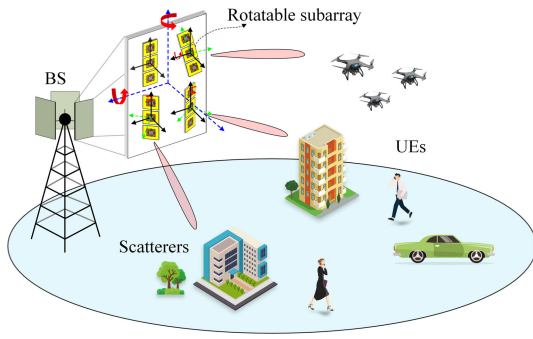


Fig. 1. Illustration of the rotatable antenna system.

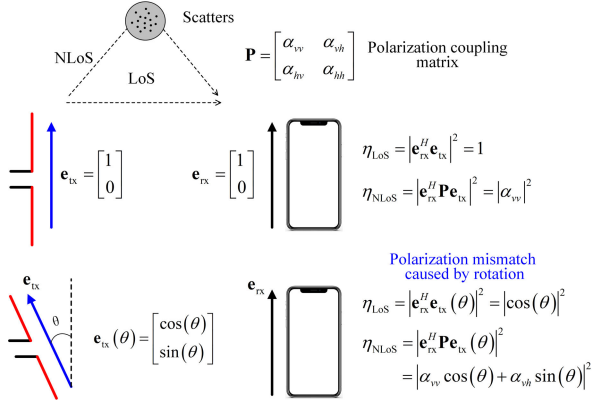


Fig. 2. Rotation-induced polarization mismatch in line-of-sight (LoS) and non-line-of-sight (NLoS) channels.

adjusting positions and rotations while adhering to mechanical constraints [22]. Recent surveys and works also highlight broader applications—such as machine-type communications, indoor localization, and space-air-ground integrated networks [23], [24], [25].

A. Motivation

Despite the promise of RAs, however, most existing works adopt overly simplified RA models: rotation is typically approximated as a change in the boresight direction or as a modulation of scalar gain patterns. Such approximations overlook critical electromagnetic (EM) effects inherent to real antenna systems:

(i) Phase distortion: when antennas rotate, the far-field phase response varies significantly with spatial direction. Ignoring this variation leads to inaccurate modeling of interference and signal combining [26].

(ii) Polarization mismatch: in dual-polarized systems, rotation alters the antenna's polarization orientation, potentially misaligning it with the dominant eigenmodes of the MIMO channel. This effect reduces the effective gain and causes link-level degradation [27]. As illustrated in Fig. 2, rotations can significantly tilt the polarization vector, leading to misalignment between transmit and receive polarizations and thereby inducing polarization-mismatch loss.

(iii) Rotation-induced mutual coupling: in compact base-station deployments, the mutual coupling between adjacent antenna modules varies with their relative orientations. These coupling changes distort embedded radiation patterns, altering both the magnitude and phase of the radiated fields [28].

Ignoring such deformation can lead to pseudo-optimal rotation decisions and inaccurate sum-rate predictions.

In short, RA systems demand a modeling framework that faithfully captures the rotation-dependent behavior of phase, polarization, and mutual coupling. Otherwise, performance evaluations and orientation optimization may diverge significantly from real-world deployments.

B. Contribution

To address these challenges, this paper proposes a unified evaluation-and-optimization framework for rotatable antennas that integrates accurate electromagnetic modeling, realistic antenna hardware, and sum-rate-driven orientation optimization. The main contributions are summarized as follows:

- First, the complex vector element pattern (CVEP) is employed in this work for the first time to accurately characterize the radiation behavior of RA elements. The CVEP inherently captures the amplitude, phase, and polarization components of the antenna radiation, and is explicitly parameterized by 3D orientation. These rotated CVEPs are then embedded into the 2×2 per-ray polarization model defined in the 3rd generation partnership project (3GPP) channel framework, enabling precise evaluation of polarization mismatch under arbitrary rotation.
- Next, a two-stage mutual coupling calibration model for RAs is proposed for the first time in this work, based on offline dictionary construction and online interpolation. In the offline stage, the proposed approach samples the CVEPs under joint rotation states of the RA subarrays to construct a rotation-dependent mutual coupling calibration dictionary. In the online stage, a power-consistent interpolation method is developed to accurately reconstruct the distorted radiation pattern under arbitrary subarray orientations.
- Furthermore, a compact dual-polarized RA element is designed and a 6×3 RA-BS prototype is assembled and measured in this work. The proposed RA element achieves a miniaturized footprint of $0.45\lambda \times 0.45\lambda \times 0.28\lambda$,¹ a high front-to-back ratio (FBR) of 16.7 dB, and stable radiation characteristics over a wide range of rotation angles within the 6.2–7.5 GHz frequency band. Both simulations and measurements of the RA-BS prototype confirm that the proposed EM model accurately captures the rotation-induced variations in gain, polarization, and radiation pattern shape.
- Finally, a joint BS-level and subarray-level orientation optimization framework is proposed in this work. We formulate a sum-rate maximization problem for RA-enabled downlink multi-user MIMO (MU-MIMO) systems, considering both BS-level and subarray-level orientations. A low-complexity alternating feasible-direction algorithm is developed to solve the constrained nonlinear problem. Numerical results confirm that the proposed model yields significantly higher sum-rates than conventional gain-only or mutual-coupling-agnostic baselines.

¹ λ denotes the wavelength corresponding to the center frequency of the antenna's operating band.

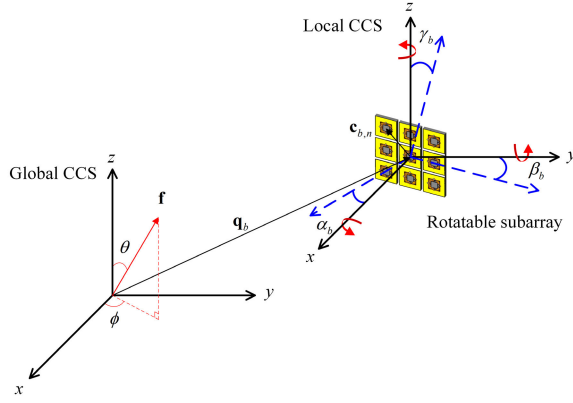


Fig. 3. Illustration of the geometry of the b -th rotatable subarray.

The rest of this paper is organized as follows. Section II presents the RA-BS system model and the phase-, polarization- and mutual-coupling-aware channel model. Section III formulates the sum-rate maximization problem and proposes an efficient orientation optimization algorithm. Section IV describes the compact RA design and validates its rotational EM characteristics. Section V provides system-level simulation results, and Section VI concludes the paper.

Notation: Letters a , $\mathbf{a}$, and $\mathbf{A}$ represent scalars, vectors, and matrices, respectively. $\mathbf{A}^T$, $\mathbf{A}^H$, $\mathbf{A}^{-1}$, $\|\mathbf{A}\|$ and $\mathbb{E}\{\mathbf{A}\}$ denote transpose, Hermitian (conjugate transpose), inverse, 2 norm, and expectation of $\mathbf{A}$, respectively. $\mathbf{A}_{i,j}$ represents the (i, j) -th element of $\mathbf{A}$. $|\cdot|$ denotes the modulus. $\mathbf{I}_d$ denotes the identity matrix of size d . $\text{diag}(\mathbf{a})$ represents a diagonal matrix with the entries of $\mathbf{a}$ on its main diagonal.

II. SYSTEM MODEL

In this section, we first establish the array geometry and orientation model of the RA BS. Furthermore, a phase-, polarization-, and mutual-coupling-aware channel model tailored for the RA BS is developed based on the rotation mapping of the CVEP.

A. RA-BS Model

As illustrated in Fig. 3, we consider a RA-enabled cellular communication system, where the BS consists of B rotatable subarrays, denoted by the set $\mathcal{B} = 1, 2, \dots, B$. Each subarray can perform 3D rotation around the x -, y -, and z -axes. The b -th rotatable subarray is composed of N antenna elements forming a uniform planar (or linear) array, represented by the set $\mathcal{N} = 1, 2, \dots, N$.

The position and orientation of the b -th subarray's rotation center are given by

$$\mathbf{q}_b = [x_b, y_b, z_b]^T, \quad (1)$$

$$\mathbf{u}_b = [\alpha_b, \beta_b, \gamma_b]^T \quad (2)$$

where α_b , β_b , and γ_b denote its rotation angles around the x -, y -, and z -axes, respectively. The position of the n -th antenna element within the b -th subarray can be expressed as

$$\mathbf{r}_{b,n}(\mathbf{q}_b, \mathbf{u}_b) = \mathbf{q}_b + \mathbf{R}(\mathbf{u}_b) \mathbf{c}_{b,n} \quad (3)$$

where $\mathbf{c}_{b,n}$ denotes the location of the n -th element's phase center relative to the rotation center in the local Cartesian coordinate system (CCS). $\mathbf{R}(\mathbf{u}_b)$ denotes the active extrinsic rotation matrix that rotates local vectors about the fixed z -, y -, and x -axes, in this order, following the right-hand rule [29]

$$\mathbf{R}(\mathbf{u}_b) = \mathbf{R}_x(\alpha_b) \mathbf{R}_y(\beta_b) \mathbf{R}_z(\gamma_b) \quad (4)$$

which yields

$$\mathbf{R}(\mathbf{u}_b) = \begin{bmatrix} c_{\beta_b} c_{\gamma_b} & -c_{\beta_b} s_{\gamma_b} & s_{\beta_b} \\ c_{\alpha_b} s_{\gamma_b} + s_{\alpha_b} s_{\beta_b} c_{\gamma_b} & c_{\alpha_b} c_{\gamma_b} - s_{\alpha_b} s_{\beta_b} s_{\gamma_b} & -s_{\alpha_b} c_{\beta_b} \\ s_{\alpha_b} s_{\gamma_b} - c_{\alpha_b} s_{\beta_b} c_{\gamma_b} & s_{\alpha_b} c_{\gamma_b} + c_{\alpha_b} s_{\beta_b} s_{\gamma_b} & c_{\alpha_b} c_{\beta_b} \end{bmatrix} \quad (5)$$

where $c_x = \cos x$ and $s_x = \sin x$ are defined for notational simplicity; $\mathbf{R}_x$, $\mathbf{R}_y$ and $\mathbf{R}_z$ can be obtained in (2.63) of [29]. This convention is adopted to match the physical mechanical rotation of the proposed RA-BS [29]. It differs from the intrinsic local-axis convention used in [30].

Based on the above geometry, we define the departure angle of the l -th path of the m -th scatterer corresponding to the k -th downlink user as $\boldsymbol{\Omega}_{m,l}^{(k)} = (\theta_{m,l}^{(k)}, \phi_{m,l}^{(k)})$, where $\theta_{m,l}^{(k)}$ and $\phi_{m,l}^{(k)}$ denote the elevation and azimuth departure angles, respectively. Accordingly, the spatial angle $\tilde{\boldsymbol{\Omega}}_{m,l}^{(k,b)}(\mathbf{u}_b) = (\tilde{\theta}_{m,l}^{(k,b)}, \tilde{\phi}_{m,l}^{(k,b)})$ on the antenna radiation pattern after rotation can be derived as

$$\tilde{\theta}_{m,l}^{(k,b)} = \arccos \left([0 \ 0 \ 1] \mathbf{R}^{-1}(\mathbf{u}_b) \mathbf{f}_{m,l}^{(k)} \right), \quad (6)$$

$$\tilde{\phi}_{m,l}^{(k,b)} = \arg \left([1 \ j \ 0] \mathbf{R}^{-1}(\mathbf{u}_b) \mathbf{f}_{m,l}^{(k)} \right) \quad (7)$$

where $\mathbf{f}_{m,l}^{(k)} = [\sin \theta_{m,l}^{(k)} \cos \phi_{m,l}^{(k)}, \sin \theta_{m,l}^{(k)} \sin \phi_{m,l}^{(k)}, \cos \theta_{m,l}^{(k)}]^T$ denotes the pointing vector corresponding to direction $(\theta_{m,l}^{(k)}, \phi_{m,l}^{(k)})$.

B. Phase- and Polarization-Aware Model

To simultaneously capture both the amplitude and phase information of antenna radiation, this section employs the CVEP to accurately characterize the far-field radiation behavior of each antenna element. Specifically, for a far-field angular direction $\boldsymbol{\Omega} = (\theta, \phi)$, the radiated field of the b -th antenna is represented as [31]

$$\mathbf{E}_b(\boldsymbol{\Omega}) = [E_r^{(b)}(\boldsymbol{\Omega}), E_\theta^{(b)}(\boldsymbol{\Omega}), E_\phi^{(b)}(\boldsymbol{\Omega})]^T \in \mathbb{C}^{3 \times 1} \quad (8)$$

where $E_r^{(b)}$, $E_\theta^{(b)}$, and $E_\phi^{(b)}$ denote the field components along the $\mathbf{r}$, $\boldsymbol{\theta}$, and $\boldsymbol{\phi}$ directions, respectively. Here, $\boldsymbol{\Omega}$ captures only the angular distribution of the radiation pattern. The distance-dependent propagation effects, including spherical spreading and propagation phase, are decoupled from the CVEP and are separately accounted for by the array phase factors and path-dependent terms in the channel model. $E_r^{(b)}(\boldsymbol{\Omega}) \equiv 0$ under the far-field assumption; the r -component is retained in (8) only for notational completeness of the spherical-basis representation and the coordinate transformation. Hence, for each direction $\boldsymbol{\Omega}$, the polarization state is fully determined by the complex tangential components $E_\theta^{(b)}(\boldsymbol{\Omega})$ and $E_\phi^{(b)}(\boldsymbol{\Omega})$

through their relative magnitudes and phases. Therefore, the CVEP preserves the amplitude, phase, and polarization characteristics of the radiated field.

For each antenna element in the BS, the corresponding CVEP can be obtained through full-wave EM simulation tools such as CST Studio [32] or by measurement. In this work, the simulated or measured embedded radiation patterns² (ERPs) are adopted as the CVEPs.

When the b -th rotatable subarray undergoes an active rotation as $\mathbf{u}_b$, the corresponding pre-rotation direction for a given observation direction is first obtained from (6) and (7), denoted by $\tilde{\boldsymbol{\Omega}}_0(\mathbf{u}_b)$. Since the local spherical bases associated with $\tilde{\boldsymbol{\Omega}}_0(\mathbf{u}_b)$ and $\boldsymbol{\Omega}_0$ are generally different, the field cannot be rotated directly in spherical coordinates. Instead, $\mathbf{E}_b(\tilde{\boldsymbol{\Omega}}_0(\mathbf{u}_b))$ is first transformed to the Cartesian basis through $\mathbf{A}(\tilde{\boldsymbol{\Omega}}_0(\mathbf{u}_b))$, then rotated by $\mathbf{R}(\mathbf{u}_b)$, and finally projected onto the spherical basis associated with $\boldsymbol{\Omega}_0$ through $\mathbf{A}^T(\boldsymbol{\Omega}_0)$. Therefore, the rotated field response toward $\boldsymbol{\Omega}_0$ is given by

$$\tilde{\mathbf{E}}_b(\boldsymbol{\Omega}_0; \mathbf{u}_b) = \mathbf{A}^T(\boldsymbol{\Omega}_0) \mathbf{R}(\mathbf{u}_b) \mathbf{A}(\tilde{\boldsymbol{\Omega}}_0(\mathbf{u}_b)) \mathbf{E}_b(\tilde{\boldsymbol{\Omega}}_0(\mathbf{u}_b)) \quad (9)$$

where

$$\mathbf{A}(\boldsymbol{\Omega}) = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \quad (10)$$

denotes the coordinate transformation matrix from the spherical coordinate system to the Cartesian coordinate system.

It can be clearly observed from (9) that rotation significantly alters the polarization state of the antenna radiation field. Accurately modeling such rotation-induced polarization variation is essential for evaluating the polarization coupling and mismatch performance of RA systems.

A typical example can be illustrated as follows: for a radiation direction $\boldsymbol{\Omega} = (\pi/2, \pi/2)$ with an original field vector $\mathbf{E}(\boldsymbol{\Omega}) = [0, 0, 1]$, when the antenna rotates $\pi/2$ about the y -axis, the field at the same direction becomes $[0, -1, 0]$, indicating a distinct change in polarization. If the rotation-induced polarization variation were neglected, one would incorrectly assume the field after rotation remains $[0, 0, 1]$, leading to a substantial error in evaluating the actual performance of the RA system.

C. Mutual-Coupling-Aware Model

In RA systems, antenna rotation not only alters the polarization orientation of the radiated field but also changes the mutual coupling between antenna elements, which significantly reshapes the embedded radiation pattern of each antenna and affects the end-to-end link gain [33].

To accurately model the rotation- and mutual-coupling-dependent behavior of the RA-BS, it is first necessary to

²The embedded radiation pattern of an antenna element is defined as the radiation pattern when that element is excited while all other antenna elements (ports) are terminated with matched loads. In this case, the mutual coupling among antenna elements is inherently included in the embedded radiation pattern [28], [31].

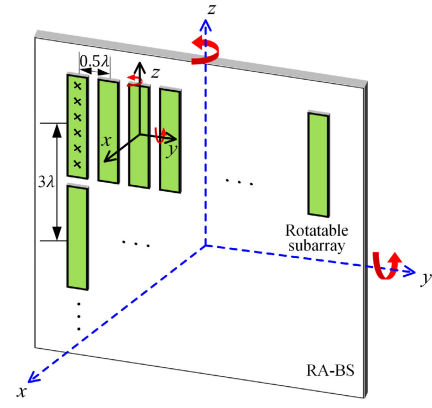


Fig. 4. 3D geometry of the RA-BS and subarray structure.

clarify the reconfigurable rotation structure of the RA-BS. Fig. 4 illustrates the BS layout adopted in this paper. To maximize spatial degrees of freedom within a fixed aperture, we adopt a conventional planar BS-array layout: the BS comprises multiple rotatable, dual-polarized 1×6 linear subarrays forming a uniform planar array. The phase-center spacing is 0.5λ along the horizontal (row, y) direction and 3λ along the vertical (column, z) direction. The BS boresight points along the x -axis. To avoid mechanical blockage, each subarray is allowed to rotate within bounded ranges about the z - and y -axes. Each subarray's six co-polarized elements are driven by an equal-amplitude, in-phase 1 to 6 power divider; hence, each subarray has two radio frequency (RF) ports corresponding to two orthogonal polarizations. For each subarray and port, we obtain the CVEP via full-wave simulation or measurement. To isolate rotational effects on communication performance, we assume the mechanical rotation center coincides with the phase center of each subarray, i.e., rotation does not translate the subarray.

Remark 1: Because the inter-row (vertical) phase-center spacing is 3λ , the rotation-induced inter-row mutual coupling can be safely neglected. In addition, for non-adjacent subarrays within the same row, the phase-center spacing ($\geq 1\lambda$) and radiator spacing ($\geq 0.5\lambda$) ensure that their rotation-induced mutual coupling variation remains weak [28], [31]. Therefore, the rotation-aware mutual coupling can be effectively simplified to interactions with the left and right nearest subarrays.

On this basis, the mutual-coupling-aware pattern can be accurately modeled as the combination of an independent rotation term and a mutual-coupling calibration term, which can be expressed as

$$\tilde{\mathbf{E}}_b^{\text{mc}}(\boldsymbol{\Omega}; \mathbf{u}_b, \mathbf{u}_{\mathcal{N}_b}) = \tilde{\mathbf{E}}_b(\boldsymbol{\Omega}; \mathbf{u}_b) + \Delta \mathbf{E}_b^{\text{mc}}(\boldsymbol{\Omega}; \mathbf{u}_b, \mathbf{u}_{\mathcal{N}_b}) \quad (11)$$

where $\Delta \mathbf{E}_b^{\text{mc}}(\boldsymbol{\Omega}; \mathbf{u}_b, \mathbf{u}_{\mathcal{N}_b})$ denotes the mutual-coupling calibration term; $\mathbf{u}_{\mathcal{N}_b} \triangleq \text{col}\{\mathbf{u}_n : n \in \mathcal{N}_b\}$ denotes the stacked orientation vector of the same-row neighboring subarrays of subarray b , and $\mathcal{N}_b = \{n \in \{b-1, b+1\} : 1 \leq n \leq RC, \lfloor \frac{n-1}{C} \rfloor = \lfloor \frac{b-1}{C} \rfloor\}$ denotes the same-row neighbor set of subarray b in an $R \times C$ BS, where $\lfloor \cdot \rfloor$ denotes the floor operator. $\Delta \mathbf{E}_b^{\text{mc}}(\boldsymbol{\Omega}) = \mathbf{0}$ when $\mathbf{u}_b = \mathbf{u}_{\mathcal{N}_b} = \mathbf{0}$. Because the mutual coupling deformation

depends on relative orientations, we introduce the joint rotation state as

$$\xi_b = [\mathbf{u}_b; \mathbf{u}_{\mathcal{N}_b}] \in \mathbb{R}^D, D = \mu(1 + |\mathcal{N}_b|) \quad (12)$$

where $\mu \in \{1, 2, 3\}$ denotes the per-subarray rotational dimensionality (e.g., z -only (1), zy (2), or full 3-axis (3)).

A closed-form expression for $\Delta \mathbf{E}_b^{\text{mc}}$ is generally intractable, whereas the corresponding CVEPs/ERPs under sampled joint states can be obtained offline by full-wave EM simulation or measurement. We therefore adopt a two-stage offline-online reconstruction strategy: a rotation-dependent mutual-coupling dictionary is first built offline, and then queried online at arbitrary joint states.

Firstly, the per-axis sampling sets can be defined as

$$\begin{aligned} \alpha &\in \mathcal{A} \\ &= \left\{ \alpha_s^{(l_\alpha)} | l_\alpha = 0, 1, \dots, L_\alpha - 1, \alpha_{\min} \leq \alpha_s^{(l_\alpha)} \leq \alpha_{\max} \right\}, \\ \beta &\in \mathcal{U} \\ &= \left\{ \beta_s^{(l_\beta)} | l_\beta = 0, 1, \dots, L_\beta - 1, \beta_{\min} \leq \beta_s^{(l_\beta)} \leq \beta_{\max} \right\}, \\ \gamma &\in \mathcal{C} \\ &= \left\{ \gamma_s^{(l_\gamma)} | l_\gamma = 0, 1, \dots, L_\gamma - 1, \gamma_{\min} \leq \gamma_s^{(l_\gamma)} \leq \gamma_{\max} \right\}. \end{aligned} \quad (13)$$

Then, the joint grid can be expressed as

$$\mathcal{X}_b = \mathcal{A}^{(b)} \times \mathcal{U}^{(b)} \times \mathcal{C}^{(b)} \times \prod_{n \in \mathcal{N}_b} \left(\mathcal{A}^{(n)} \times \mathcal{U}^{(n)} \times \mathcal{C}^{(n)} \right) \quad (14)$$

with multi-index $\mathbf{m}$ labeling grid points. The sampled joint states can be represented as

$$\begin{aligned} \xi_b^{(\mathbf{m})} &\in \mathcal{X}_b, \\ \mathbf{m} \in \mathcal{I} &\triangleq \left\{ 0, \dots, L_\alpha^{(1)} - 1 \right\} \times \dots \times \left\{ 0, \dots, L_\gamma^{1+|\mathcal{N}_b|} - 1 \right\}. \end{aligned} \quad (15)$$

For each $\xi_b^{(\mathbf{m})}$, the ground-truth mutual-coupling-calibration field is extracted as

$$\Delta \mathbf{E}_b^{\text{mc}}(\Omega; \xi_b^{(\mathbf{m})}) = \tilde{\mathbf{E}}_b^{\text{mc}}(\Omega; \xi_b^{(\mathbf{m})}) - \tilde{\mathbf{E}}_b(\Omega; \mathbf{u}_b^{(\mathbf{m})}). \quad (16)$$

For an arbitrary queried state $\xi_b^{(\mathbf{m})}$, we reconstruct the calibration field by natural-neighbor interpolation (NNI) [34], i.e.,

$$\Delta \mathbf{E}_b^{\text{mc}}(\Omega; \xi_b) \approx \sum_{\mathbf{m} \in \mathcal{I}} w_{\mathbf{m}}(\xi_b) \Delta \mathbf{E}_b^{\text{mc}}(\Omega; \xi_b^{(\mathbf{m})}) \quad (17)$$

where $w_{\mathbf{m}}$ denotes the weighting coefficient; $w_{\mathbf{m}}(\xi_b) \geq 0$ and $\sum_{\mathbf{m} \in \mathcal{I}} w_{\mathbf{m}}(\xi_b) = 1$. By substituting (17) into (11), the mutual-coupling-aware radiation field of the rotatable antenna can be derived.

Direct interpolation of complex vector fields may introduce a slight drift in the total radiated power. To remove this artifact, we impose a scalar power-consistency correction. Specifically, the target radiated power is interpolated as

$$P_{\text{rad}}^{\text{tar}}(\xi_b) \approx \sum_{\mathbf{m} \in \mathcal{I}} w_{\mathbf{m}}^p(\xi_b) P_{\text{rad}}(\xi_b^{(\mathbf{m})}) \quad (18)$$

whereas the total radiated power of the raw reconstructed field is

$$\hat{P}_{\text{rad}}(\xi_b) = \int_{4\pi} \frac{|\hat{E}_{b,\theta}^{\text{mc}}(\Omega; \xi_b)|^2 + |\hat{E}_{b,\phi}^{\text{mc}}(\Omega; \xi_b)|^2}{2\eta_0} d\Omega \quad (19)$$

where η_0 denotes the free-space wave impedance. Hence, the corresponding power-calibration factor can be obtained as

$$\kappa_b(\xi_b) = \sqrt{\frac{P_{\text{rad}}^{\text{tar}}(\xi_b)}{\hat{P}_{\text{rad}}(\xi_b)}} \quad (20)$$

and the final power-consistent field is given by

$$\tilde{\mathbf{E}}_b^{\text{mc,pc}}(\Omega; \xi_b) = \kappa_b(\xi_b) \tilde{\mathbf{E}}_b^{\text{mc}}(\Omega; \xi_b). \quad (21)$$

Since $\kappa_b(\xi_b)$ is a real-valued scalar, this correction restores the total radiated power without altering the relative phase relations or polarization composition of the reconstructed field. In implementation, $\kappa_b(\xi_b)$ depends only on the queried joint state and is therefore precomputed offline on a dense auxiliary grid and stored as a scalar lookup table. During the online optimization, $\kappa_b(\xi_b)$ is obtained by direct lookup or simple linear interpolation, so the spherical integration in (19) is not executed online.

D. Joint BS-Level and Subarray-Level Rotation Model

To further expand the spatial DoF of the BS, we incorporate the overall rotational capability of the BS into the RA-BS model. As illustrated in Fig. 4, in addition to the independent rotation of each subarray, the BS is assumed to rotate around the global x -, y -, and z -axes. The BS orientation is defined as

$$\mathbf{t} = [\alpha_{\text{bs}}, \beta_{\text{bs}}, \gamma_{\text{bs}}]^T. \quad (22)$$

First, the overall rotation of the BS changes the phase center of each subarray. The phase-center position of the b -th subarray under BS rotation can be expressed as

$$\mathbf{r}_b(\mathbf{t}) = \mathbf{R}(\mathbf{t}) \mathbf{q}_b. \quad (23)$$

Furthermore, because the overall BS rotation preserves the relative orientations among subarrays, it does not introduce additional mutual-coupling variation. Therefore, the BS-level rotation acts on the already calibrated subarray field through the same pre-rotation lookup, Cartesian-basis rotation, and spherical-basis reprojection mechanism as in (9), with $\tilde{\mathbf{E}}_b^{\text{mc,pc}}(\tilde{\Omega}; \xi_b)$ serving as the input field. Under this condition, the radiation pattern of the b -th subarray after BS-level rotation can be expressed as

$$\tilde{\mathbf{E}}_b^{\text{bs,sa}}(\Omega; \xi_b, \mathbf{t}) = \mathbf{A}^T(\Omega) \mathbf{R}(\mathbf{t}) \mathbf{A}(\tilde{\Omega}) \tilde{\mathbf{E}}_b^{\text{mc,pc}}(\tilde{\Omega}; \xi_b) \quad (24)$$

where $\tilde{\Omega}$ can be obtained from (6) and (7) with Ω as the input.

Based on the above, the mapping between the rotation angles and the antenna radiation field under both subarray-level and BS-level rotations can be comprehensively modeled. It is noteworthy that the proposed formulation consistently incorporates the rotation-dependent effects of phase, polarization, and mutual coupling in the resulting radiation field.

E. Channel Model

In this subsection, we embed the power-calibrated, phase-, polarization-, and mutual-coupling-consistent patterns derived in (24) into a wideband, geometry-based stochastic channel model compliant with the 3GPP 38.901 [30]. The construction preserves the small-scale fading and cluster powers specified by 3GPP while exposing explicit dependence on the RA joint orientations.

Consider a BS comprising B rotatable subarrays; each subarray provides two RF ports (orthogonal polarizations), so the BS has $N_t = 2B$ Tx ports. Each user is equipped with N_r receive (Rx) ports. The channel is represented by L clusters, each with S subpaths. For the (ℓ, s) -th ray, denote the transmit departure unit vector with angle of departure (AoD) $\Omega_{\ell,s}^{\text{TX}} = (\theta_{\ell,s}^{\text{TX}}, \phi_{\ell,s}^{\text{TX}})$ as

$$\hat{\mathbf{s}}_{\ell,s}^{\text{TX}} = [\sin \theta_{\ell,s}^{\text{TX}} \cos \phi_{\ell,s}^{\text{TX}}, \sin \theta_{\ell,s}^{\text{TX}} \sin \phi_{\ell,s}^{\text{TX}}, \cos \theta_{\ell,s}^{\text{TX}}]^T. \quad (25)$$

Similarly, denote the receive arrival unit vector as $\hat{\mathbf{s}}_{\ell,s}^{\text{RX}}$ with angle of arrival (AoA) $\Omega_{\ell,s}^{\text{RX}} = (\theta_{\ell,s}^{\text{RX}}, \phi_{\ell,s}^{\text{RX}})$. Denote $\tau_{\ell,s}$ and $\nu_{\ell,s}$ as the excess delay and Doppler frequency, respectively, and denote $P_{\ell,s}$ as the per-ray power with $\sum_s P_{\ell,s} = P_\ell$ and $\sum_\ell P_\ell = 1$. Since the radial component of CVEP is identically zero in the far field, we simplify the CVEP of subarray b as

$$\tilde{\mathbf{E}}_b^{\text{bs},sa}(\Omega; \mathbf{u}, \mathbf{t}) = [\tilde{E}_{b,\theta}^{\text{bs},sa}(\Omega; \mathbf{u}, \mathbf{t}), \tilde{E}_{b,\phi}^{\text{bs},sa}(\Omega; \mathbf{u}, \mathbf{t})]^T \in \mathbb{C}^{2 \times 1} \quad (26)$$

where $\mathbf{u} = [\mathbf{u}_1; \mathbf{u}_2; \dots; \mathbf{u}_B] \in \mathbb{R}^{3B \times 1}$ stacks all BS orientations. Based on this, stacking all subarrays (two ports per subarray) yields the Tx block as

$$\mathbf{F}^{\text{TX}}(\Omega; \mathbf{u}, \mathbf{t}) = [\tilde{\mathbf{E}}_1^{\text{bs},sa}(\Omega; \mathbf{u}, \mathbf{t}), \dots, \tilde{\mathbf{E}}_{N_t}^{\text{bs},sa}(\Omega; \mathbf{u}, \mathbf{t})] \in \mathbb{C}^{2 \times N_t}. \quad (27)$$

On the Rx side, $\mathbf{F}^{\text{RX}}(\Omega) \in \mathbb{C}^{2 \times N_r}$ stacks the N_r -port CVEPs similarly. With Tx subarray phase-centers $\{\mathbf{r}_b\}$ and Rx port locations $\{\mathbf{r}_q^{\text{RX}}\}$, the array phase factors for ray (ℓ, s) can be expressed as

$$a_b^{\text{TX}}(\mathbf{r}_b) = \exp(-jk_0 \hat{\mathbf{s}}_{\ell,s}^{\text{TX}} \cdot \mathbf{r}_b), \quad (28)$$

$$a_q^{\text{RX}}(\mathbf{r}_q^{\text{RX}}) = \exp(jk_0 \hat{\mathbf{s}}_{\ell,s}^{\text{RX}} \cdot \mathbf{r}_q^{\text{RX}}) \quad (29)$$

where $k_0 = 2\pi f/c$ denotes the wavenumber at frequency f and c denotes the speed of light. On this basis, the sub-channel between the BS and the k -th user of ray (ℓ, s) at time t_τ and frequency f can be derived as [30]

$$\begin{aligned} \mathbf{H}_k^{(\ell,s)}(\mathbf{u}, \mathbf{t}; f, t_\tau) \\ = \sqrt{P_{\ell,s}} \mathbf{A}_{\ell,s}^{\text{RX}} \mathbf{F}_{k,\ell,s}^{\text{RX}} (\Omega_{\ell,s}^{\text{RX}})^T \mathbf{X}_{\ell,s} \mathbf{F}_{\ell,s}^{\text{TX}} (\Omega_{\ell,s}^{\text{TX}}; \mathbf{u}, \mathbf{t}) \mathbf{A}_{\ell,s}^{\text{TX}} \\ \cdot \exp(-j2\pi f \tau_{\ell,s}) \exp(j2\pi \nu_{\ell,s} t_\tau) \end{aligned} \quad (30)$$

where

$$\mathbf{A}_{\ell,s}^{\text{TX}} = \text{diag}(a_1^{\text{TX}}(\mathbf{r}_1), \dots, a_{N_t}^{\text{TX}}(\mathbf{r}_{N_t})) \in \mathbb{C}^{N_t \times N_t}, \quad (31)$$

$$\mathbf{A}_{\ell,s}^{\text{RX}} = \text{diag}(a_1^{\text{RX}}(\mathbf{r}_1^{\text{RX}}), \dots, a_{N_r}^{\text{RX}}(\mathbf{r}_{N_r}^{\text{RX}})) \in \mathbb{C}^{N_r \times N_r} \quad (32)$$

and $\mathbf{X}_{\ell,s} \in \mathbb{C}^{2 \times 2}$ denotes the per-ray polarization coupling matrix, which can be expressed as

$$\mathbf{X}_{\ell,s} = \begin{bmatrix} \exp(j\Phi_{\ell,s}^{\theta\theta}) & \sqrt{\kappa_{1,\ell,s}^{-1}} \exp(j\Phi_{\ell,s}^{\theta\phi}) \\ \sqrt{\kappa_{2,\ell,s}^{-1} \chi_{\ell,s}^{-1}} \exp(j\Phi_{\ell,s}^{\phi\theta}) & \sqrt{\chi_{\ell,s}^{-1}} \exp(j\Phi_{\ell,s}^{\phi\phi}) \end{bmatrix} \quad (33)$$

where $\Phi_{\ell,s}^{\theta\theta}$, $\Phi_{\ell,s}^{\theta\phi}$, $\Phi_{\ell,s}^{\phi\theta}$ and $\Phi_{\ell,s}^{\phi\phi}$ denote the initial phase of θ -to- θ , ϕ -to- θ , θ -to- ϕ , and ϕ -to- ϕ polarization, respectively; $\kappa_{1,\ell,s}$ and $\kappa_{2,\ell,s}$ denote the cross-polarization ratios (XPRs), and $\chi_{\ell,s}$ denotes the co-polarization ratio (CPR). For simplicity, we denote $\mathbf{H}_k^{(\ell,s)}(\mathbf{u}, \mathbf{t}; f, t)$ as $\mathbf{H}_k^{(\ell,s)}(\mathbf{u}, \mathbf{t})$.

Summing over all clusters and subpaths, the channel between the BS and the k -th user can be expressed as

$$\mathbf{H}_k(\mathbf{u}, \mathbf{t}) = \sum_{\ell=1}^L \sum_{s=1}^S \mathbf{H}_k^{(\ell,s)}(\mathbf{u}, \mathbf{t}). \quad (34)$$

The proposed model captures and characterizes the effects of rotation-induced variations in phase, polarization, and mutual coupling that arise from both the BS-level and subarray-level rotations on the wireless channel.

III. ROTATION OPTIMIZATION ALGORITHM

In this section, based on the phase-, polarization-, and mutual-coupling-aware model, we develop a joint BS-level and subarray-level orientation optimization framework tailored for RA-BSs to maximize the achievable sum-rate (ASR) in MU-MIMO systems.

A. Signal Model

Consider a downlink MU-MIMO communication system with K user equipment (UE). The received signal of the k -th UE can be expressed as

$$\mathbf{y}_k = \mathbf{G}_k \mathbf{H}_k(\mathbf{u}, \mathbf{t}) \mathbf{W} \mathbf{s} + \mathbf{G}_k \mathbf{n}_k = \bar{\mathbf{H}}_k(\mathbf{u}, \mathbf{t}) \mathbf{W} \mathbf{s} + \bar{\mathbf{n}}_k \quad (35)$$

where $\mathbf{G}_k$ denotes the receiver combining matrix, $\mathbf{W}$ is the transmit precoding matrix, $\mathbf{n}_k \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_{N_r})$ represents the additive complex white Gaussian noise (AWGN), and the effective noise term satisfies $\bar{\mathbf{n}}_k \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{G}_k \mathbf{G}_k^H)$. The equivalent channel is defined as

$$\bar{\mathbf{H}}_k(\mathbf{u}, \mathbf{t}) \triangleq \mathbf{G}_k \mathbf{H}_k(\mathbf{u}, \mathbf{t}) \in \mathbb{C}^{d_k \times N_t} \quad (36)$$

where d_k denotes the number of data streams assigned to the k -th UE, and the total number of data streams is denoted as $d \triangleq \sum_{k=1}^K d_k$. The transmitted symbol is defined as

$$\mathbf{s} = [\mathbf{s}_1^T \dots, \mathbf{s}_K^T]^T \in \mathbb{C}^{d \times 1} \quad (37)$$

with $\mathbf{E}\{\mathbf{s}\mathbf{s}^H\} = \mathbf{I}_d$ and $\mathbf{s}_k \in \mathbb{C}^{d_k \times 1}$. In this paper, we consider the singular value decomposition (SVD)-based receiver to mitigate inter-stream interference and the zero-forcing (ZF) precoder to suppress inter-user interference. Specifically, the channel of the k -th UE can be decomposed as

$$\mathbf{H}_k(\mathbf{u}, \mathbf{t}) = \mathbf{U}_k \Sigma_k \mathbf{V}_k^H \quad (38)$$

where $\mathbf{U}_k \in \mathbb{C}^{N_r \times N_r}$ and $\mathbf{V}_k \in \mathbb{C}^{N_t \times N_t}$ are unitary matrices, and $\Sigma_k = \text{diag}(\sigma_{k,1}, \dots, \sigma_{k,M}) \in \mathbb{C}^{N_r \times N_t}$ with

$\sigma_{k,1} \geq \sigma_{k,2} \geq \dots \geq 0$, $M = \min(N_r, N_t)$. The receiver combining matrix for the k -th UE is defined as

$$\mathbf{G}_k \triangleq \mathbf{U}_k(:, 1:d_k)^H \in \mathbb{C}^{d_k \times N_r}. \quad (39)$$

Thus, the effective noise term satisfies $\bar{\mathbf{n}}_k \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_{d_k})$. To design the ZF precoder, we first stack all UE channels into a composite channel matrix

$$\mathbf{H}_{\text{eff}}(\mathbf{u}, \mathbf{t}) \triangleq [\bar{\mathbf{H}}_1^T(\mathbf{u}, \mathbf{t}), \dots, \bar{\mathbf{H}}_K^T(\mathbf{u}, \mathbf{t})]^T \in \mathbb{C}^{d \times N_t}. \quad (40)$$

The ZF precoder is then given by

$$\mathbf{W}_{\text{ZF}} = \mathbf{H}_{\text{eff}}^H(\mathbf{u}, \mathbf{t}) (\mathbf{H}_{\text{eff}}(\mathbf{u}, \mathbf{t}) \mathbf{H}_{\text{eff}}^H(\mathbf{u}, \mathbf{t}))^{-1}. \quad (41)$$

Each column vector $\mathbf{w}_i$ of $\mathbf{W}_{\text{ZF}}$ is normalized to ensure equal transmit power per stream, yielding

$$\mathbf{W} = \sqrt{\frac{P_{\text{tx}}}{d}} [\tilde{\mathbf{w}}_1, \dots, \tilde{\mathbf{w}}_d], \tilde{\mathbf{w}}_i = \frac{\mathbf{w}_i}{\|\mathbf{w}_i\|_2} \quad (42)$$

where P_{tx} denotes the total transmit power. Accordingly, the signal-to-interference-plus-noise ratio (SINR) of the i -th stream of the k -th UE is given by

$$\text{SINR}_{k,i}(\mathbf{u}, \mathbf{t}) = \frac{\left| \mathbf{g}_{k,i} \mathbf{H}_k(\mathbf{u}, \mathbf{t}) \mathbf{w}_{f_{k,i}} \right|^2}{\sum_{l=1, (l,n) \neq (k,i)}^K \sum_{n=1}^{d_l} \left| \mathbf{g}_{k,i} \mathbf{H}_k(\mathbf{u}, \mathbf{t}) \mathbf{w}_{f_{l,n}} \right|^2 + \frac{\sigma^2}{P_{\text{tx}}}} \quad (43)$$

where $\mathbf{g}_{k,i}$ denotes the i -th row of $\mathbf{G}_k$ and $f_{a,b} = \sum_{k=1}^{a-1} d_k + b$.

The achievable rate of the k -th UE can be expressed as

$$R_k(\mathbf{u}, \mathbf{t}) = \sum_{i=1}^{d_k} \log_2(1 + \text{SINR}_{k,i}(\mathbf{u}, \mathbf{t})) \quad (44)$$

and the ASR of the MU-MIMO system can be obtained as

$$R_{\text{sum}}(\mathbf{u}, \mathbf{t}) = \sum_{k=1}^K R_k(\mathbf{u}, \mathbf{t}). \quad (45)$$

B. Achievable Sum-Rate Maximization Problem

In order to maximize the ASR of the MU-MIMO system, we propose a joint BS-level and subarray-level orientation optimization framework. The ASR maximization problem can be formulated as

$$\begin{aligned} (\text{P1}): \quad & \max_{\mathbf{u}, \mathbf{t}} R_{\text{sum}}(\mathbf{u}, \mathbf{t}) \\ \text{s.t.} \quad & \mathbf{u}_L \leq \mathbf{u} \leq \mathbf{u}_U \\ & \mathbf{t}_L \leq \mathbf{t} \leq \mathbf{t}_U. \end{aligned} \quad (46)$$

The box constraints in (P1) ensure that the mechanical rotation angles of the BS and each subarray remain within physically feasible limits imposed by the RA-BS hardware design.

Problem (P1) is a nonconvex bound-constrained maximization problem, where the coupling between the BS orientation and the subarray orientations renders direct joint optimization computationally prohibitive. To address this challenge, we adopt an alternating-optimization (AO) strategy that iteratively updates the two orientation blocks in a cyclic manner while keeping the other fixed.

At the i -th iteration, given the current pair $(\mathbf{u}^{(i)}, \mathbf{t}^{(i)})$, the AO procedure alternates between the following two subproblems

$$(\text{P1-t}): \max_{\mathbf{t}} R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}) \quad \text{s.t.} \quad \mathbf{t}_L \leq \mathbf{t} \leq \mathbf{t}_U, \quad (47)$$

$$(\text{P1-u}): \max_{\mathbf{u}} R_{\text{sum}}(\mathbf{u}, \mathbf{t}^{(i)}) \quad \text{s.t.} \quad \mathbf{u}_L \leq \mathbf{u} \leq \mathbf{u}_U. \quad (48)$$

Subproblem (P1-t) optimizes the global rotation of the BS while keeping the subarray orientations fixed, whereas subproblem (P1-u) refines the local rotations of all subarrays based on the updated BS orientation. Both subproblems involve only simple box constraints, making them well suited to feasible-direction (projected gradient-ascent) methods with guaranteed monotonic ascent [35].

C. Feasible-Direction-Based Problem Solvers

In the proposed solver, the ascent directions are evaluated by coordinate-wise forward finite differences on the coded objective. This choice is deliberate rather than merely for implementation convenience. Specifically, the objective is not a simple closed-form function of the rotation variables, but a nested composite map through the orientation-dependent EM-aware channel reconstruction, the effective-channel assembly, the ZF precoder recomputation, and the SVD-based receiver update. Although derivatives of individual algebraic blocks can in principle be derived separately, a complete end-to-end analytical gradient would be lengthy, strongly implementation-dependent, and considerably less transparent than a direct objective-based evaluation. We therefore adopt forward finite differences as the exact algorithmic reference in this work, and introduce a low-complexity batched surrogate in Section III-D to improve scalability for large-array settings.

Let $\Pi_{[\mathbf{a}, \mathbf{b}]}(\cdot)$ denote the elementwise projection onto the interval $[\mathbf{a}, \mathbf{b}]$. The gradients of $R_{\text{sum}}(\mathbf{u}, \mathbf{t})$ with respect to $\mathbf{t}$ and $\mathbf{u}$, denoted $\nabla_{\mathbf{t}} R_{\text{sum}}$ and $\nabla_{\mathbf{u}} R_{\text{sum}}$, can be obtained by

$$\begin{aligned} & [\nabla_{\mathbf{t}} R_{\text{sum}}(\mathbf{u}, \mathbf{t})]_j \\ &= \lim_{\varepsilon \rightarrow 0} \frac{R_{\text{sum}}(\mathbf{u}, \mathbf{t} + \varepsilon \mathbf{e}_t^j) - R_{\text{sum}}(\mathbf{u}, \mathbf{t})}{\varepsilon}, 1 \leq j \leq 3, \end{aligned} \quad (49)$$

$$\begin{aligned} & [\nabla_{\mathbf{u}} R_{\text{sum}}(\mathbf{u}, \mathbf{t})]_j \\ &= \lim_{\varepsilon \rightarrow 0} \frac{R_{\text{sum}}(\mathbf{u} + \varepsilon \mathbf{e}_u^j, \mathbf{t}) - R_{\text{sum}}(\mathbf{u}, \mathbf{t})}{\varepsilon}, 1 \leq j \leq 3B \end{aligned} \quad (50)$$

where $\mathbf{e}_t^j \in \mathbb{R}^{3 \times 1}$ and $\mathbf{e}_u^j \in \mathbb{R}^{3B \times 1}$ are vectors with a one as the j -th element and zeros elsewhere. On this basis, given $\mathbf{u}^{(i)}$, the feasible-direction step for solving (P1-t) can be expressed as

$$\mathbf{t}^{(i+1)} = \Pi_{[\mathbf{t}_L, \mathbf{t}_U]} \left(\mathbf{t}^{(i)} + \eta_t \nabla_{\mathbf{t}} R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i)}) \right) \quad (51)$$

where $\eta_t > 0$ denotes the initial trial line-search coefficient in the projected feasible-direction update. It is an algorithmic quantity rather than a physical angular increment. The actual accepted BS-level update is determined only after Armijo

backtracking and box projection. η_t is used in Armijo backtracking line search [36] to guarantee a sufficient ascent, i.e.

$$R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i+1)}) \geq R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i)}) + \sigma_t \eta_t \left\| \nabla_{\mathbf{t}} R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i)}) \right\|_2^2 \quad (52)$$

where $\sigma_t \in (0, 1)$ denotes the Armijo constant.

Algorithm 1 Alternating Feasible-Direction Algorithm for Solving (P1)

Input: $\mathbf{t}^{(0)} \in [\mathbf{t}_L, \mathbf{t}_U]$, $\mathbf{u}^{(0)} \in [\mathbf{u}_L, \mathbf{u}_U]$, η_t , η_u , σ_t , σ_u , $I_{\text{max}}, \varepsilon_c$

Output: $(\mathbf{u}^*, \mathbf{t}^*)$, R_{sum}^*

- 1: Compute $R_{\text{sum}}(\mathbf{u}^{(0)}, \mathbf{t}^{(0)})$
- 2: **for** $i = 0$ to $I_{\text{max}} - 1$ **do**
- 3: $\mathbf{g}_t^{(i,i)} \leftarrow \nabla_{\mathbf{t}} R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i)})$
- 4: Update $\mathbf{t}^{(i+1)}$ by (51)
- 5: **while** $R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i+1)}) < R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i)}) + \sigma_t \eta_t \left\| \mathbf{g}_t^{(i,i)} \right\|_2^2$ **do**
- 6: $\eta_t \leftarrow \sigma_t \eta_t$, update $\mathbf{t}^{(i+1)}$ by (51)
- 7: **end while**
- 8: $\mathbf{g}_u^{(i,i+1)} \leftarrow \nabla_{\mathbf{u}} R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i+1)})$
- 9: Update $\mathbf{u}^{(i+1)}$ by (53)
- 10: **while** $R_{\text{sum}}(\mathbf{u}^{(i+1)}, \mathbf{t}^{(i+1)}) < R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i+1)}) + \sigma_u \eta_u \left\| \mathbf{g}_u^{(i,i+1)} \right\|_2^2$ **do**
- 11: $\eta_u \leftarrow \sigma_u \eta_u$, update $\mathbf{u}^{(i+1)}$ by (53)
- 12: **end while**
- 13: **if** $|R_{\text{sum}}(\mathbf{u}^{(i+1)}, \mathbf{t}^{(i+1)}) - R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i)})| \leq \varepsilon_c$ **then**
- 14: **break**
- 15: **end if**
- 16: **end for**
- 17: **return:** $\mathbf{u}^* = \mathbf{u}^{(i+1)}$, $\mathbf{t}^* = \mathbf{t}^{(i+1)}$, $R_{\text{sum}}^* = R_{\text{sum}}(\mathbf{u}^{(i+1)}, \mathbf{t}^{(i+1)})$

Given the updated BS orientation $\mathbf{t}^{(i+1)}$, the feasible-direction step for solving (P1-u) can be expressed as

$$\mathbf{u}^{(i+1)} = \Pi_{[\mathbf{u}_L, \mathbf{u}_U]} \left(\mathbf{u}^{(i)} + \eta_u \nabla_{\mathbf{u}} R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i+1)}) \right) \quad (53)$$

where the step size $\eta_u > 0$ is determined analogously to (52) that satisfies

$$R_{\text{sum}}(\mathbf{u}^{(i+1)}, \mathbf{t}^{(i+1)}) \geq R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i+1)}) + \sigma_u \eta_u \left\| \nabla_{\mathbf{u}} R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i+1)}) \right\|_2^2. \quad (54)$$

The alternating feasible-direction algorithm proceeds by repeatedly executing (47)–(54) until convergence. Specifically, it terminates when the ASR increment between two successive outer iterations falls below a small threshold, i.e.

$$\left| R_{\text{sum}}(\mathbf{u}^{(i+1)}, \mathbf{t}^{(i+1)}) - R_{\text{sum}}(\mathbf{u}^{(i)}, \mathbf{t}^{(i)}) \right| \leq \varepsilon_c. \quad (55)$$

The optimization framework for solving problem (P1) is summarized in Algorithm 1.

D. Low-Complexity Batched Gradient Evaluation

The subarray-level gradient in (50) is defined by conventional coordinate-wise finite differences and remains the exact algorithmic reference in this paper. To reduce the repeated EM-aware channel reconstructions in the subarray-level update, we further introduce a low-complexity batched gradient-evaluation scheme by exploiting the local dependence structure of the adopted EM-aware channel model.

Let the EM-aware channel reconstructed at the current iterate $(\mathbf{u}, \mathbf{t})$ be partitioned into Tx-subarray channel blocks as

$$\mathbf{H}^E(\mathbf{u}, \mathbf{t}) = [\mathbf{H}_1^E(\mathbf{u}, \mathbf{t}), \mathbf{H}_2^E(\mathbf{u}, \mathbf{t}), \dots, \mathbf{H}_B^E(\mathbf{u}, \mathbf{t})] \quad (56)$$

where $\mathbf{H}_b^E(\mathbf{u}, \mathbf{t})$ denotes the Tx-channel block associated with the b -th rotatable subarray. For a fixed subarray-level rotation DoF index $l \in \{1, \dots, \mu_{\text{sba}}\}$, where $\mu_{\text{sba}} \in \{1, 2, 3\}$ denotes the per-subarray rotation DoFs, let $\mathbf{e}_u^{b,l}$ denote the canonical basis vector corresponding to the l -th DoF of subarray b .

The exact perturbed channel required by the finite-difference evaluation in (50) is

$$\mathbf{H}_{(b,l)}^E \triangleq \mathbf{H}^E(\mathbf{u} + \varepsilon \mathbf{e}_u^{b,l}, \mathbf{t}) = [\mathbf{H}_{(b,l),1}^E, \mathbf{H}_{(b,l),2}^E, \dots, \mathbf{H}_{(b,l),B}^E]. \quad (57)$$

Under the local-mutual-coupling model in Section II-C, the j -th Tx-channel block depends only on the local queried state $\xi_j = [\mathbf{u}_j; \mathbf{u}_{\mathcal{N}_j}]$. Hence, perturbing subarray b can affect only those blocks whose local states explicitly contain $\mathbf{u}_b$. Define the corresponding affected-block set as

$$\mathcal{B}_b \triangleq \{j \in \{1, \dots, B\} : b \in \{j\} \cup \mathcal{N}_j\}. \quad (58)$$

Hence, we can obtain $\mathbf{H}_{(b,l),j}^E = \mathbf{H}_j^E(\mathbf{u}, \mathbf{t})$ for $j \notin \mathcal{B}_b$.

To amortize this cost across all B subarrays for the l -th DoF direction, we define the jointly perturbed direction as $\mathbf{d}_u^{(l)} \triangleq \sum_{b=1}^B \mathbf{e}_u^{b,l}$ and reconstruct only one jointly perturbed channel

$$\mathbf{H}_{J,l}^E \triangleq \mathbf{H}^E(\mathbf{u} + \varepsilon \mathbf{d}_u^{(l)}, \mathbf{t}) = [\mathbf{H}_{J,l,1}^E, \mathbf{H}_{J,l,2}^E, \dots, \mathbf{H}_{J,l,B}^E]. \quad (59)$$

Based on (56)–(59), we construct the batched surrogate channel for the perturbation of subarray b along DoF l as

$$\hat{\mathbf{H}}_{(b,l)}^E \triangleq [\hat{\mathbf{H}}_{(b,l),1}^E, \hat{\mathbf{H}}_{(b,l),2}^E, \dots, \hat{\mathbf{H}}_{(b,l),B}^E] \quad (60)$$

with

$$\hat{\mathbf{H}}_{(b,l),j}^E = \begin{cases} \mathbf{H}_{J,l,j}^E, & j = b, \\ \mathbf{H}_j^E(\mathbf{u}, \mathbf{t}), & j \neq b. \end{cases} \quad (61)$$

That is, the b -th perturbed block is refreshed from the jointly perturbed channel, whereas other blocks are directly reused from the baseline channel.

In the decoupled rigid-rotation reference case, each Tx-channel block depends only on the local subarray orientation itself, and thus $\mathcal{B}_b = \{b\}$. In this case,

$$\hat{\mathbf{H}}_{(b,l)}^E = \mathbf{H}_{(b,l)}^E \quad (62)$$

which shows that the batched construction is exactly equivalent to the conventional coordinate-wise perturbed reconstruction.

In the mutual-coupling-aware case, however, the batched surrogate in (61) generally differs from the exact perturbed

channel in (57), because the refreshed self block is extracted from the jointly perturbed channel in (59), while the neighboring affected blocks are directly reused from the baseline channel. We therefore define the resulting reconstruction discrepancy as

$$\mathbf{E}_{(b,l)}^E(\varepsilon) \triangleq \mathbf{H}_{(b,l)}^E - \hat{\mathbf{H}}_{(b,l)}^E. \quad (63)$$

By the definition of $\mathcal{B}_b$ and the adopted local-mutual-coupling model, one has

$$\mathbf{E}_{(b,l),j}^E(\varepsilon) = \mathbf{0}, \quad j \notin \mathcal{B}_b \quad (64)$$

that is, the discrepancy is strictly local and is supported only on the affected block set $\mathcal{B}_b$.

For $j = b$, let the self block be written as a local EM-aware map

$$\mathbf{H}_b^E = \mathcal{F}_b(\boldsymbol{\xi}_b, \mathbf{t}). \quad (65)$$

Let $\boldsymbol{\xi}_b^{(b,l)}(\varepsilon)$ and $\bar{\boldsymbol{\xi}}_b^{(l)}(\varepsilon)$ denote the local queried states induced by the exact single perturbation and the jointly perturbed construction, respectively. Then

$$\mathbf{H}_{(b,l),b}^E = \mathcal{F}_b(\boldsymbol{\xi}_b^{(b,l)}(\varepsilon), \mathbf{t}), \quad \hat{\mathbf{H}}_{(b,l),b}^E = \mathcal{F}_b(\bar{\boldsymbol{\xi}}_b^{(l)}(\varepsilon), \mathbf{t}). \quad (66)$$

If $\mathcal{F}_b$ is differentiable with respect to $\boldsymbol{\xi}_b$ in a neighborhood of the current iterate, then it is locally Lipschitz continuous there. Hence, for sufficiently small ε , there exists a constant $L_{b,l}^{\text{self}} > 0$ such that

$$\|\mathbf{E}_{(b,l),b}^E(\varepsilon)\|_F \leq L_{b,l}^{\text{self}} \|\boldsymbol{\xi}_b^{(b,l)}(\varepsilon) - \bar{\boldsymbol{\xi}}_b^{(l)}(\varepsilon)\|_2 \leq L_{b,l}^{\text{self}} |\varepsilon|. \quad (67)$$

For each neighboring affected block $j \in \mathcal{B}_b \setminus \{b\}$, let $\mathbf{H}_j^E = \mathcal{F}_j(\boldsymbol{\xi}_j, \mathbf{t})$. Since the batched surrogate directly reuses the baseline block for such j , one has

$$\mathbf{H}_{(b,l),j}^E = \mathcal{F}_j(\boldsymbol{\xi}_j^{(b,l)}(\varepsilon), \mathbf{t}), \quad \hat{\mathbf{H}}_{(b,l),j}^E = \mathcal{F}_j(\boldsymbol{\xi}_j, \mathbf{t}). \quad (68)$$

If $\mathcal{F}_j$ is differentiable with respect to $\boldsymbol{\xi}_j$ in a neighborhood of the current iterate, then for sufficiently small ε , there exists a constant $L_{b,l,j}^{\text{nbr}} > 0$ such that

$$\|\mathbf{E}_{(b,l),j}^E(\varepsilon)\|_F \leq L_{b,l,j}^{\text{nbr}} \|\boldsymbol{\xi}_j^{(b,l)}(\varepsilon) - \boldsymbol{\xi}_j\|_2 \leq L_{b,l,j}^{\text{nbr}} |\varepsilon|, \quad j \in \mathcal{B}_b \setminus \{b\}. \quad (69)$$

Combining (64), (67), and (69), there exists a constant $L_b > 0$ such that

$$\|\mathbf{E}_{(b,l)}^E(\varepsilon)\|_F \leq L_b |\varepsilon|. \quad (70)$$

This shows that, under the local-mutual-coupling model, the batched substitution introduces only a local channel reconstruction discrepancy, and this discrepancy shrinks to zero as $\varepsilon \rightarrow 0$. The proposed batched scheme should thus be understood as a locality-based low-complexity surrogate for the exact coordinate-wise perturbed objective evaluation in the mutual-coupling-aware case. Its main purpose is to substantially reduce the repeated EM-aware channel reconstructions in the subarray-level update, thereby improving the scalability of the proposed framework for large arrays.

E. Complexity Analysis

The computational cost of the proposed solver is dominated by repeated evaluations of $R_{\text{sum}}(\mathbf{u}, \mathbf{t})$ during numerical-gradient computation and Armijo backtracking. We decompose one objective evaluation into two parts

$$C_{\text{obj}} = C_{\text{ch}} + C_{\text{tr}} \quad (71)$$

where C_{ch} denotes the cost of one online EM-aware channel reconstruction, and C_{tr} denotes the cost of transceiver recomputation and ASR evaluation once the channel is available.

The offline stage requires N_I full-wave EM solves to construct the dictionary and thus incurs a one-time cost

$$C_{\text{off}} = N_I C_{\text{EM}} \quad (72)$$

where C_{EM} denotes the cost of one EM solve. This cost is hardware-dependent but is completely decoupled from the online AO iterations.

For the considered dual-polarized subarray architecture, one has $N_t = 2B$. Under the proposed EM-aware framework, C_{ch} consists of online pattern interpolation and channel assembly, yielding

$$C_{\text{ch}} = \mathcal{O}(N_t N_{\text{ray}} \log N_I + N_t N_{\text{ray}} N_r) \quad (73)$$

where N_{ray} denotes the total number of queried rays in channel construction. The cost C_{tr} is dominated by per-user receive-side SVD, ZF precoder recomputation, and the subsequent ASR evaluation, which can be summarized as

$$C_{\text{tr}} = \mathcal{O}(K N_t N_r^2 + d^2 N_t + d^3) \quad (74)$$

where d denotes the total number of scheduled data streams. Hence, for fixed K , d , and N_r , both C_{ch} and C_{tr} scale linearly with B .

If the subarray-level gradient were evaluated by a standard coordinate-wise forward difference, then one subarray-level update would require $\mu_{\text{sba}} B$ singly perturbed objective evaluations. Its complexity would therefore be

$$C_{\text{naive,u}} = \mu_{\text{sba}} B (C_{\text{ch}} + C_{\text{tr}}). \quad (75)$$

Since C_{ch} itself already scales linearly with B , the EM-model-specific part of (75) grows quadratically with B , namely,

$$\mathcal{O}(\mu_{\text{sba}} B^2 N_{\text{ray}} (\log N_I + N_r)) \quad (76)$$

which becomes the dominant bottleneck for large arrays.

To reduce this repeated EM-aware reconstruction cost, we employ the batched gradient-evaluation scheme introduced in Section III-D. For each subarray-level DoF direction, only one jointly perturbed channel needs to be reconstructed, while the baseline channel at the current iterate is directly reused from the accepted objective evaluation. The remaining perturbed objectives are then formed by local block substitution and incur the corresponding transceiver/ASR evaluation cost. Up to the lower-order cost of local block substitution, which is absorbed into C_{ch} , the resulting subarray-level complexity is

$$C_{\text{batch,u}} = \mu_{\text{sba}} C_{\text{ch}} + \mu_{\text{sba}} B C_{\text{tr}}. \quad (77)$$

Therefore, the EM-model-specific part is reduced from the quadratic scaling in (76) to

$$\mathcal{O}(\mu_{\text{sba}}BN_{\text{ray}}(\log N_I + N_r)) \quad (78)$$

that is, from quadratic to linear growth in B .

Finally, let $\bar{I}_{\text{bt}}^t$ and $\bar{I}_{\text{bt}}^u$ denote the average additional objective evaluations caused by Armijo backtracking in the BS-level and subarray-level updates, respectively, and let $\mu_{\text{bs}} \in \{1, 2, 3\}$ denote the number of BS-level rotation DoFs. Ignoring the common baseline objective evaluation shared by all perturbed gradients at the same iterate, the per-outer-iteration complexity of the exact-gradient-based solver and the batch-gradient-based solver can be summarized as

$$C_{\text{outer}}^{\text{exact}} = (\mu_{\text{sba}}B + \mu_{\text{bs}} + \bar{I}_{\text{bt}}^t + \bar{I}_{\text{bt}}^u)C_{\text{ch}} + (\mu_{\text{sba}}B + \mu_{\text{bs}} + \bar{I}_{\text{bt}}^t + \bar{I}_{\text{bt}}^u)C_{\text{tr}} \quad (79)$$

and

$$C_{\text{outer}}^{\text{batch}} = (\mu_{\text{sba}} + \mu_{\text{bs}} + \bar{I}_{\text{bt}}^t + \bar{I}_{\text{bt}}^u)C_{\text{ch}} + (\mu_{\text{sba}}B + \mu_{\text{bs}} + \bar{I}_{\text{bt}}^t + \bar{I}_{\text{bt}}^u)C_{\text{tr}} \quad (80)$$

respectively.

From the above analysis, the EM-model-specific overhead of the proposed batched subarray-gradient evaluation scales linearly with the number of rotatable subarrays B . This favorable scaling substantially improves the computational scalability of the proposed framework for large-array settings.

IV. ROTATABLE ANTENNA ARRAY PROTOTYPE

A critical challenge in designing the RA lies in enabling wide-angle subarray rotation while avoiding collisions between adjacent subarrays for densely packed arrays (with element spacing $\leq \lambda/2$). This size constraint requires antenna miniaturization with reflector size of each subarray less than $\lambda/2$. Simultaneously, good impedance matching ($\text{VSWR} < 1.5$) and port isolation (> 15 dB) must be maintained throughout the rotational process without severe performance degradation. However, conventional BS antennas use relatively large reflectors (typically 0.8λ [37]). The edge diffraction energy is increased with narrower reflector, causing higher backward radiation and degraded FBR, elevating system-level interference in cellular communication networks. Consequently, it is necessary to achieve antenna miniaturization with FBR enhancement.

A. Compact Rotatable BS Antenna Design

To address these challenges, a compact RA BS element suitable for large-angle rotation is designed in this work based on the current path extension principle [38]. In addition, a vertical comb-shaped metallic fence is employed to ensure a high FBR [39].

The geometry of the proposed BS antenna is shown in Fig. 5. The proposed antenna consists of a pair of $\pm 45^\circ$ loop-dipoles, Y-shaped feeding structures, four vertical plates with meandering lines, a parasitic patch, and a metallic reflector. As shown in Fig. 5(b), the Y-shaped feeding structures and the loop-dipoles are printed on the top and bottom layers of a 0.254 mm thick WL-CT440 substrate (with a relative

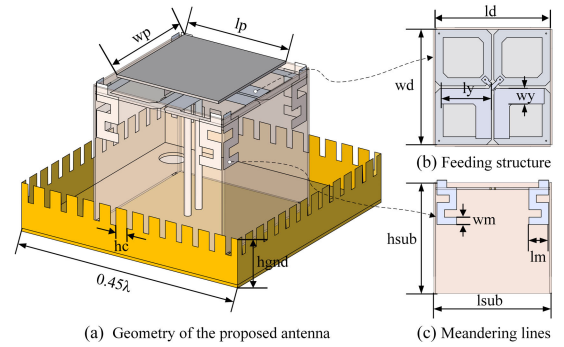


Fig. 5. Geometry of the proposed RA BS antenna.

TABLE I
DETAILED DIMENSIONS OF THE PROPOSED ANTENNA (UNIT: MM)

Variable	Value	Variable	Value
lp	9.0	hsub	9.2
wp	9.0	lsub	10.0
ld	11.0	lm	1.8
wd	11.0	wm	0.7
wy	1.6	ly	5.0
hc	1.5	hgnd	3.7

permittivity of 4.38 and a loss tangent of 0.005), respectively. Eight meandering lines (printed on a 0.508 mm thick WL-CT350 substrate with a relative permittivity of 3.66 and a loss tangent of 0.0039) extend downward from the four corners of the loop-dipoles, which enable antenna miniaturization to $0.26\lambda \times 0.26\lambda \times 0.28\lambda$, as shown in Fig. 5(c). A parasitic patch (printed on a 0.254 mm thick F4BM220 substrate with a relative permittivity of 2.2 and a loss tangent of 0.001) positioned 1 mm above the loop-dipole substrate enhances broadband impedance matching. The four edges of the metallic reflector are folded upward and shaped into comb-shaped corrugations to improve the antenna FBR. The final size of the reflector is $0.45\lambda \times 0.45\lambda \times 0.07\lambda$, enabling $\pm 60^\circ$ rotation without collision for half-wavelength-spaced arrays. The detailed parameters of the proposed antenna are listed in Table I.

The proposed antenna achieves broadband impedance matching across 6.2–7.5 GHz ($S_{11} < -15$ dB) and high port isolation higher than 30 dB. It exhibits a gain of 7.1 dBi, a high FBR of 16.7 dB, and a cross-polarization discrimination (XPD) greater than 20 dB, indicating a stable and well-shaped radiation pattern. On this basis, a 6×1 subarray composed of the proposed elements is excited by a 1-to-6 equal-amplitude in-phase power divider, serving as the rotatable unit of the RA-BS.

B. Electromagnetic Characteristics of Rotatable Antenna Array

For half-wavelength inter-element spacing, the embedded response of each subarray port is dominated by its nearest neighbors [28], [31]. Full-wave simulations on a manageable 6×3 proxy array can accurately capture this mutual coupling locality while being computationally affordable. Therefore, a 6×3 block is adopted to extract orientation-dependent embedded radiation patterns representative of larger panels. According to the mutual-coupling-aware model in

TABLE II

ROTATION STATES AND CORRESPONDING MIDDLE-ELEMENT GAIN VARIATION FOR THE 1×3 ARRAY (SYMMETRY-REDUCED CASES)

Middle-element rotation category	Joint rotation state	Configuration example	Middle-element gain (dBi)
No rotation	[0, 0, 0]	, ,	5.74
	[1, 0, 0]	/, ,	6.45
	[2, 0, 0]	\, ,	6.22
	[1, 0, 1]	/, , /	7.52
	[2, 0, 1]	\, , /	7.36
	[1, 0, 2]	/, , \	7.45
Clockwise rotation	[0, 1, 0]	, /,	5.02
	[1, 1, 0]	/, /,	4.90
	[2, 1, 0]	\, /,	4.81
	[0, 1, 1]	, /, /	4.94
	[1, 1, 1]	/, /, /	4.82
	[2, 1, 1]	\, /, /	4.81
	[0, 1, 2]	, /, \	3.76
	[1, 1, 2]	/, /, \	3.66
	[2, 1, 2]	\, /, \	3.62

Section II-C, constructing a calibration dictionary via full-wave EM simulations requires sampling a 6-D joint rotation space, which is computationally prohibitive. To overcome this, we exploit the rotational characteristics of the RA-BS to reduce the sampling dimensionality and thus lower the dictionary construction complexity.

Firstly, in cellular deployments, the UE layout and the small elevation angular spread confine link sensitivity to a narrow elevation sector around the downtilt direction [30]. Therefore, as a trade-off between performance and complexity, the subarray-level elevation optimization can be simplified to the BS-level, which significantly reduces the complexity of constructing the mutual-coupling calibration dictionary while causing only a minor performance loss for the RA-BS system.

On this basis, to comprehensively analyze the effect of z -axis rotation on radiation characteristics, we enumerate all possible combinations in which the central subarray remains fixed, rotates clockwise, or rotates counter-clockwise. This yields 27 configurations, which can be reduced to 15 unique cases by exploiting structural symmetry. Owing to computational constraints, the simulation uses a 1×3 array rather than a 6×3 block, as it sufficiently captures the essential mutual coupling behavior for z -axis rotation.

Table II lists the 15 distinct rotation states (symmetry-reduced) for the 1×3 array. For simplicity, the middle-element gain is used to approximate the variation in mutual coupling: cases exhibiting larger gain deviation are considered to have stronger mutual coupling variation. In Table II, state “0” denotes no rotation, while “1” and “2” represent $+60^\circ$ and -60° around the z -axis, respectively (assuming a per-subarray rotation range of $[-60^\circ, 60^\circ]$).

From Table II, it can be observed that when the central element is fixed, its radiation pattern is mainly affected by the rotation states of its adjacent elements. Therefore, the sampling dimensionality of the mutual-coupling calibration dictionary can be reduced to two dimensions (and to one dimension for edge elements). When the central element itself rotates, the mutual coupling variation depends primarily on its own

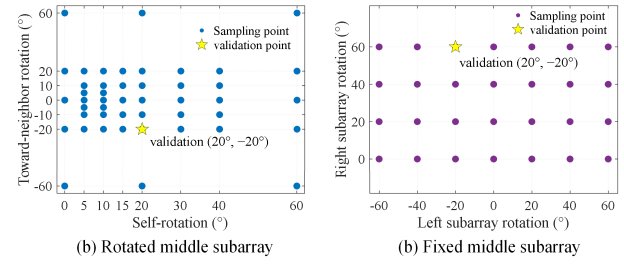


Fig. 6. Diagram of the sampling point.

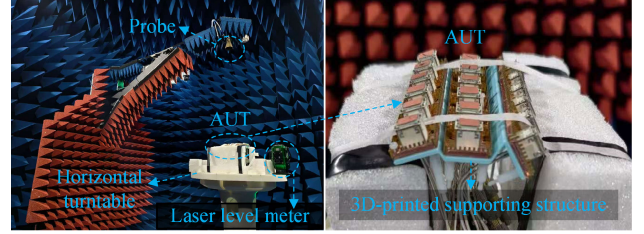


Fig. 7. Measurement photo and zoom view of the antenna under test (AUT).

rotation angle and the rotation of the neighboring element facing it, while the influence of the element on the opposite side can be negligible. Hence, in this case, the dictionary sampling can also be reduced to a two-dimensional grid.

Based on the above analysis, the original 3-D EM-simulation sampling can be reduced to two 2-D sampling grids corresponding to the cases where the central subarray rotates or remains fixed, which further significantly reduces the complexity of constructing the mutual-coupling calibration dictionary.

C. Simulation and Measurement Validation

To validate the predictive accuracy of the proposed online reconstruction framework, we select representative off-grid joint rotation states for which no direct full-wave samples are included in the offline dictionary. The validation points are marked by the yellow stars in Fig. 6. We first conducted full-wave simulations and measurements for a representative joint rotation state of $[(0^\circ, 0^\circ, 0^\circ), (0^\circ, 0^\circ, 20^\circ), (0^\circ, 0^\circ, -20^\circ)]$. The constructed sampling grid is also illustrated in Fig. 6. The natural-neighbor interpolation scheme [34] is adopted in this paper. The measurement photo is shown in Fig. 7. For simplicity, we denote the joint rotation state as $[0^\circ, 20^\circ, -20^\circ]$. During measurements, six ports of the power divider were connected to the subarray, while the remaining ports were terminated with 50Ω matched loads. Each subarray was mounted on a 3D-printed supporting structure corresponding to its rotation angle to emulate mechanical rotation.

Figs. 8(a) and (b) compare the proposed model, the direct CST full-wave simulation, and measured results of the middle subarray for the joint rotation state of $[0^\circ, 20^\circ, -20^\circ]$. Note that the curves labeled “Proposed model” and “CST simulation” correspond to two different computational processes. The “Proposed model” is the online reconstruction from the pre-computed dictionary, whereas the “CST simulation” is the direct full-wave electromagnetic solution at the same target rotation state and serves as a high-fidelity reference.

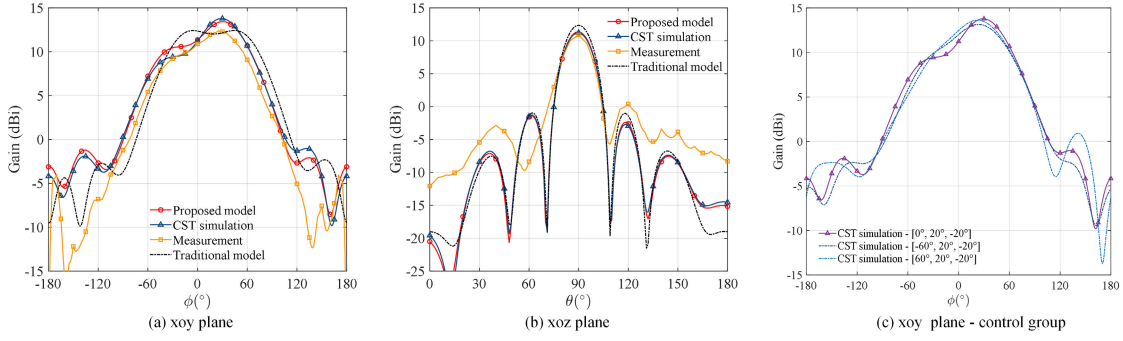


Fig. 8. Comparison and validation of the proposed mutual-coupling-aware model under the joint rotation state $[0^\circ, 20^\circ, -20^\circ]$.

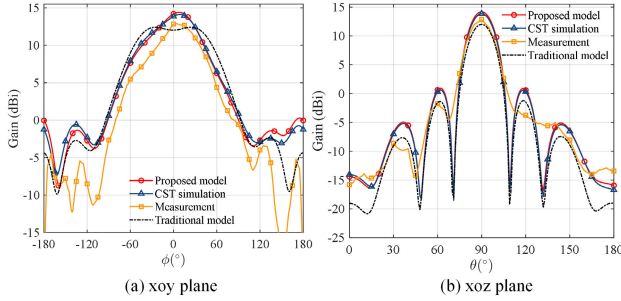


Fig. 9. Comparison and validation of the proposed mutual-coupling-aware model under the joint rotation state $[-20^\circ, 0^\circ, 60^\circ]$.

As shown in Fig. 8, compared with the ideal independent rotation, the actual rotation of the RA significantly alters the radiation pattern shape and gain, making its accurate evaluation essential for RA-BS system design, deployment, and optimization. The strong agreement among the three in *xoy* plane verifies that the proposed mutual-coupling-aware framework accurately captures the rotation-induced variations in the antenna radiation field. In the *xoz* plane, the modeled radiation pattern agrees well with the CST full-wave simulation. However, the measured pattern shows a slight deviation from both. This discrepancy mainly arises from the commercial 1-to-8 power divider, which introduces an amplitude imbalance of ± 0.4 dB and a phase imbalance of $\pm 5^\circ$ among its output ports.

To further highlight the advantage of the proposed model over the conventional rigid-body rotation scheme that neglects mutual coupling effects, Fig. 8 also presents the pattern predicted by the traditional model (black curve). As observed, the proposed framework more precisely reproduces the electromagnetic performance variations caused by rotation, thereby enabling more accurate orientation optimization of the RA-BS in practical deployments.

In addition, CST comparisons for $\pm 60^\circ$ z-axis rotations of the left subarray—i.e., $[-60^\circ, 20^\circ, -20^\circ]$ and $[60^\circ, 20^\circ, -20^\circ]$ —show negligible influence on the overall radiation pattern when the central subarray rotates toward the right, as shown in Fig. 8(c). This indicates that the opposite-side rotation has negligible impact on the mutual coupling and the radiated field.

For the case where the central subarray remains fixed, simulations and measurements were performed of the middle

subarray for the joint rotation state $[-20^\circ, 0^\circ, 60^\circ]$. Fig. 9 presents the corresponding proposed model, the direct CST simulation, and measured radiation patterns. The excellent consistency again validates the accuracy of the proposed mutual-coupling-aware model in capturing rotation-induced mutual coupling variations. In addition, the discrepancies observed in the *xoz* plane measurement are also mainly attributed to amplitude and phase imbalance of the commercial power divider.

Overall, the simulation and measurement results confirm the effectiveness and accuracy of the proposed model in characterizing rotation-induced mutual coupling variations in the RA-BS system, demonstrating a substantial improvement in pattern modeling accuracy over conventional approaches.

V. SIMULATION RESULTS

In this section, simulations are conducted to evaluate the ASR performance of downlink MU-MIMO systems based on the proposed phase-, polarization-, and mutual-coupling-aware model and the ASR-maximization orientation-optimization algorithm.

A. Simulation Setup and Numerical Robustness

Specifically, the BS employs the 1×6 dual-polarized rotatable subarray designed in Section IV, arranged in a 4×4 layout, yielding 32 active ports in total. Each UE consists of two orthogonal half-wavelength dipole pairs (four receive antennas) and operates at 6.7 GHz. Eighteen UEs are distributed within a 120° horizontal sector, at distances of 50-200 m from the BS. The BS height is set to 20 m with a mechanical down-tilt of 9° , while UE heights are uniformly distributed between 1.5 m and 22.5 m. Simulations follow the 3GPP 38.901 urban micro (UMi) non-line-of-sight (NLoS) channel model [30].

Considering the trade-off among performance, modeling accuracy, and complexity, this work adopts a jointly programmable rotation architecture at both the BS level and the subarray level. Specifically, at the BS level, the entire BS can rotate about the *z*-axis within $\pm 60^\circ$ and about the *y*-axis within $\pm 30^\circ$. At the subarray level, each subarray can rotate about its own *z*-axis within $\pm 60^\circ$. The detailed parameters are summarized in Table III. All rotation variables are represented in degrees for physical interpretability. η_t and η_u are the initial trial line-search coefficients and are adaptively reduced

TABLE III
SIMULATION PARAMETERS

Symbol	Description	Value
B	Number of rotatable subarrays in BS	16
N_t	Number of active ports in BS	32
K	Number of UE	18
N_r	Number of receive antennas of each UE	4
d_k	Number of streams scheduled for each UE	2
$\mathbf{t}_U$	Upper bound of BS overall rotation	$[0^\circ, 30^\circ, 60^\circ]$
$\mathbf{t}_L$	Lower bound of BS overall rotation	$[0^\circ, -30^\circ, -60^\circ]$
$\mathbf{u}_U^{(b)}$	Upper bound of each subarray rotation	$[0^\circ, 0^\circ, 60^\circ]$
$\mathbf{u}_L^{(b)}$	Lower bound of each subarray rotation	$[0^\circ, 0^\circ, -60^\circ]$
N_{drop}	Number of channel drops	100
σ^2	Average noise power	-80 dBm
I_{max}	Maximum number of alternating optimizations	20
η_t, η_u	Initial trial line-search coefficient	200
σ_t, σ_u	Armijo constant	0.5
ε_c	Convergence threshold	10^{-3}

during Armijo backtracking; They are not physical rotation increments.

Two user-distribution scenarios are considered: (i) a uniformly distributed case and (ii) a hotspot case. In the hotspot scenario, UEs are concentrated within a 20° azimuthal sector centered randomly inside the cell, with a radial span of 20 m. To ensure statistical convergence, the hotspot center is uniformly shifted over the cell area across channel realizations.

For comparison, six schemes under the same array configuration are simulated:

- **Fixed BS:** all rotations are fixed with $\mathbf{t} = [0^\circ, 9^\circ, 0^\circ]$ and $\mathbf{u}^{(b)} = [0^\circ, 0^\circ, 0^\circ]$.
- **BS-only RA-BS:** The subarray relative orientations are fixed at $\mathbf{u}^{(b)} = [0^\circ, 0^\circ, 0^\circ]$, and only the global BS orientation $\mathbf{t}$ is optimized. This scheme emulates conventional panel-level azimuth/downtilt optimization within the considered mechanical-orientation framework.
- **Subarray-only RA-BS (EG/LG):** The global BS orientation is fixed at $\mathbf{t} = [0^\circ, 9^\circ, 0^\circ]$, and the independent subarray orientations $\mathbf{u}$ are optimized. EG denotes the exact coordinate-wise gradient-evaluation scheme, and LG denotes the low-complexity batched gradient-evaluation scheme;
- **Gain-only RA-BS:** the antenna pattern is modeled as a magnitude-only vector field $\widehat{\mathbf{E}}_b(\Omega) = \left[\left| E_r^{(b)}(\Omega) \right|, \left| E_\theta^{(b)}(\Omega) \right|, \left| E_\phi^{(b)}(\Omega) \right| \right]^T$, neglecting rotation-induced polarization and mutual coupling changes.
- **Mutual-coupling-neglected (MC-neglected) RA-BS:** phase and polarization effects are modeled, but rotation-induced mutual-coupling variation is ignored.
- **Proposed Joint RA-BS (EG/LG):** The proposed comprehensive framework that integrates the fully accurate EM model (accounting for phase, polarization, and mutual coupling) with the joint optimization of both BS-level $\mathbf{t}$ and subarray-level $\mathbf{u}$ rotations.

In addition to ZF, regularized ZF (RZF) is also considered as a robustness baseline. Since the effective channel is normalized with the large-scale fading factored out, the regularization term

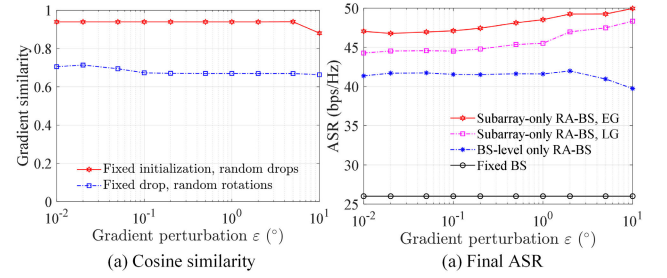


Fig. 10. Gradient fidelity and resulting ASR performance versus perturbation magnitude under uniformly distributed UEs, 25 dBm transmit power, ZF precoding, and 100 random realizations.

is weighted by the per-user large-scale fading gain

$$\mathbf{W}_{\text{RZF}} = \mathbf{H}_{\text{eff}}^H (\mathbf{H}_{\text{eff}} \mathbf{H}_{\text{eff}}^H + \mathbf{\Lambda})^{-1} \quad (81)$$

where $\mathbf{\Lambda} = \text{blkdiag}(\delta_1 \mathbf{I}_{d_1}, \dots, \delta_K \mathbf{I}_{d_K})$, $\delta_k = d\sigma^2 / (P_{\text{tx}} \zeta_k)$ and ζ_k denotes the large-scale fading gain of the k -th user. The columns of $\mathbf{W}_{\text{RZF}}$ are then normalized as in the ZF case.

To select the perturbation magnitude ε , we evaluate both the gradient fidelity and the resulting optimization performance under a representative setting with uniformly distributed UEs, 25 dBm transmit power, ZF precoding, and 100 random realizations. Fig. 10(a) plots the cosine similarity between the exact single-perturbation gradient $\mathbf{g}^{\text{ex}}$ and the proposed batched gradient $\mathbf{g}^{\text{bat}}$, defined as

$$\rho(\mathbf{g}^{\text{ex}}, \mathbf{g}^{\text{bat}}) = \frac{(\mathbf{g}^{\text{ex}})^T \mathbf{g}^{\text{bat}}}{\|\mathbf{g}^{\text{ex}}\|_2 \|\mathbf{g}^{\text{bat}}\|_2}. \quad (82)$$

The similarity remains high for small perturbations, indicating that the batched scheme preserves the dominant ascent direction of the exact gradient well. When ε becomes too large, the directional mismatch becomes more pronounced.

Fig. 10(b) further compares the final ASR under different perturbation values. The batched scheme incurs only a limited ASR loss relative to the exact gradient evaluation while still providing stable gain over the non-optimized baseline. Based on the tradeoff between gradient fidelity and final ASR, both the BS-level and subarray-level perturbations are set to 1° in the subsequent simulations. This value provides a good balance between optimization robustness and computational efficiency. The exact coordinate-wise finite-difference solver remains the theoretical reference, whereas the batched implementation serves as a scalable low-complexity surrogate.

B. Convergence Behavior and Line-Search Overhead

To further evaluate the convergence behavior of the proposed alternating feasible-direction framework, Fig. 11 plots the average ASR versus the outer-iteration index for the BS-only, subarray-only (EG/LG), and proposed joint BS-subarray (EG/LG) schemes at 25 dBm, averaged over 100 channel realizations.

As observed, all schemes exhibit stable convergence with rapid ASR improvement in the first several iterations and clear saturation afterward. In both the uniform and hotspot UE distributions, the proposed joint schemes converge to the

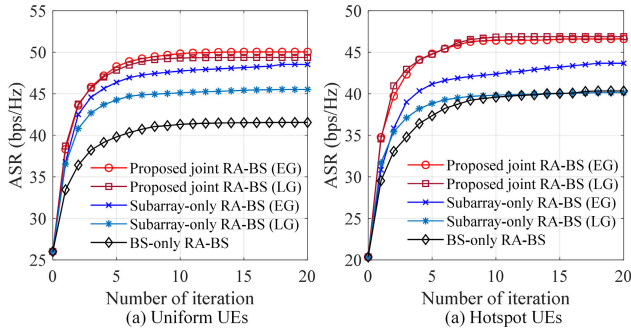


Fig. 11. Convergence behavior of the BS-only, subarray-only, and proposed joint RA-BS schemes at 25 dBm, averaged over 100 channel realizations.

TABLE IV

AVERAGE NUMBER OF ARMIJO BACKTRACKING STEPS PER ITERATION FOR THE BS-LEVEL AND SUBARRAY-LEVEL UPDATES

Scheme	Uniform		Hotspot	
	BS-level	Subarray-level	BS-level	Subarray-level
BS-only	4.16	—	4.05	—
Subarray-only (LG)	—	4.74	—	4.74
Subarray-only (EG)	—	4.77	—	4.77
Prop. joint (LG)	3.32	4.73	3.32	4.69
Prop. joint (EG)	3.33	4.71	3.31	4.71

highest stationary point. The BS-only scheme converges with the lowest optimization burden, whereas the subarray-only schemes require slightly more iterations, particularly for the EG implementation. Nevertheless, the proposed joint schemes still stabilize within a moderate number of iterations, indicating that the added optimization depth remains computationally manageable while yielding the highest ASR.

To quantify the line-search overhead, Table IV reports the average number of Armijo backtracking steps per iteration for the BS-level and subarray-level updates. The backtracking counts remain limited for all compared schemes, showing that the monotonic improvement is not achieved at the expense of excessive line-search overhead. In addition, the EG and LG schemes exhibit nearly identical backtracking behavior, indicating that the proposed batched implementation mainly reduces the gradient-evaluation cost while preserving the dominant ascent direction of the exact gradient well.

C. ASR Performance Under ZF and RZF Precoding

Fig. 12 compares the ASR performance of the BS-only, subarray-only, and proposed joint RA-BS schemes under ZF and RZF precoding for the uniform and hotspot UE distributions. Under ZF, the average ASR gains over the Fixed BS baseline in the uniform UE distribution are -9.11%, 33.11%, 51.62%, 66.13%, 75.39%, 78.32% and 81.32% for the gain-only, MC-neglected, BS-only, subarray-only (LG), subarray-only (EG), proposed joint (LG), and proposed joint (EG) schemes, respectively. In the hotspot UE distribution, the corresponding gains increase to -6.41%, 56.17%, 85.91%, 87.93%, 99.81%, 113.40% and 114.01%. These results confirm that incorporating antenna phase, polarization, and rotation-dependent mutual coupling into rotation-aware

TABLE V

REPRESENTATIVE ONLINE RUNTIMES AND ASR GAINS UNDER THE UNIFORM UE DISTRIBUTION WITH ZF PRECODING

Scheme	Runtime per drop (s)	Runtime per SNR (min)	ASR gain
Fixed BS	0.27	0.45	0%
Gain-only	15.99	26.65	-9.11%
MC-neglected	17.02	28.37	33.11%
BS-only RA-BS	11.11	18.52	51.62%
Subarray (LG)	18.97	31.61	66.13%
Subarray (EG)	40.40	67.33	75.39%
Joint (LG)	23.87	39.78	78.32%
Joint (EG)	41.46	69.10	81.32%

modeling and optimization is essential for obtaining reliable performance improvement in RA-BS systems.

Comparing the BS-only and subarray-only schemes further reveals the roles of the two rotational control layers. The BS-level rotation provides coarse global alignment and is therefore more beneficial in the hotspot scenario, where the users are spatially concentrated. In contrast, the subarray-level rotation provides finer spatial adaptation and yields more pronounced gains in the uniform UE case, as well as in the higher-SNR interference-limited regime of the hotspot case. In all scenarios, the proposed joint RA-BS schemes achieve the highest ASR, showing that the two control layers provide complementary gains and that the added optimization complexity is meaningful. Moreover, the LG implementation consistently achieves stable performance improvement with only a small loss relative to the EG implementation, which validates the effectiveness of the proposed low-complexity batched gradient scheme. Finally, the same performance ordering is preserved under RZF precoding, confirming that the benefit of the proposed EM-consistent RA-BS framework is not specific to ZF and remains applicable under a more robust linear precoder.

D. Runtime Performance

Table V reports representative online runtimes under the uniform UE distribution with ZF precoding, including the average runtime per channel realization (per drop) and per SNR point (100 drops). The offline full-wave EM dictionary construction is excluded. All runtimes are measured on the same platform (Intel i9-14900K CPU, 64 GB RAM, MATLAB 2023a) using the same implementation and stopping criteria as those adopted for the performance evaluation.

As shown, Fixed BS has the smallest runtime, while BS-only RA-BS is the fastest among the adaptive schemes due to its low-dimensional control space. By contrast, schemes involving subarray-level optimization are more time-consuming because of the repeated EM-aware channel reconstructions and gradient evaluations. The proposed low-complexity gradient (LG) scheme significantly reduces the runtime of the subarray-only and joint schemes while preserving most of the ASR gain. In particular, for the proposed joint scheme, the runtime is reduced from 41.46 s/drop to 23.87 s/drop, whereas the average ASR gain over Fixed BS decreases only from 81.32% to 78.32%. These results verify

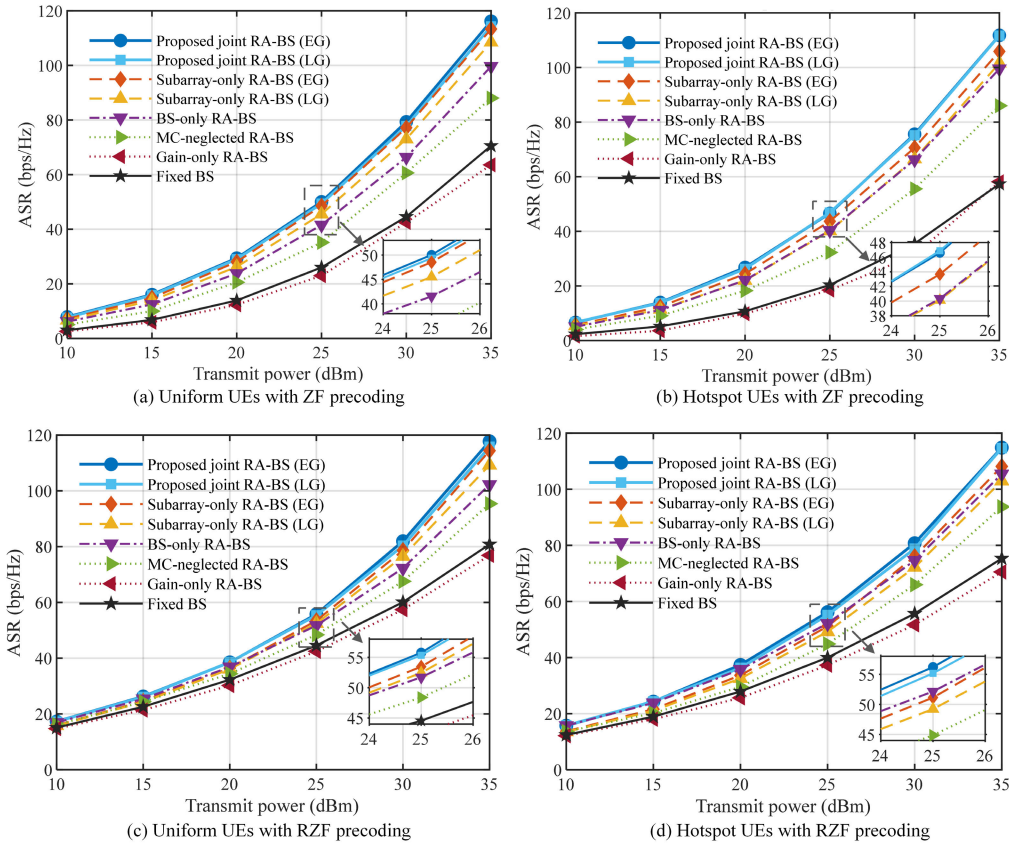


Fig. 12. ASR versus transmit power for different RA-BS schemes under ZF and RZF precoding for the uniform and hotspot UE distributions.

that the proposed joint RA-BS (LG) provides a favorable performance-complexity tradeoff.

E. Discussion on Scope, Generalization, and Practical Deployment

The proposed EM-aware framework is particularly relevant to compact dual-polarized rotatable subarrays used for orientation optimization, where mechanical rotation changes not only the boresight gain, but also the radiation phase, polarization response, and local mutual coupling. As shown in Fig. 12, simplified gain-only or mutual-coupling-neglected models yield substantially weaker gains and may even incur negative average gain in the uniform-UE case, indicating that oversimplified models may mislead the communication-level orientation decisions rather than merely causing a small modeling error.

The adopted reduced-dimensional reconstruction is physically motivated by the spatial locality of mutual coupling and is further stabilized by natural-neighbor interpolation and power-consistency correction. Its validity is therefore tied to regular array deployments with sufficiently separated phase centers, whereas more compact tightly coupled arrays would require a higher-dimensional local state. Although the specific offline dictionary values are geometry- and frequency-dependent, the proposed framework itself is generic: for a new array geometry or operating band, one only needs to regenerate the offline dictionary, while the online reconstruction and optimization procedures remain unchanged.

The dominant storage overhead comes from the offline mutual-coupling calibration dictionary rather than from the online interpolation or the power-consistency correction. In the considered regular-array setting, dictionary sharing across representative subarray classes avoids constructing an independent dictionary for every physical subarray index. For the adopted $1^\circ(\theta, \phi)$ sampling grid, one joint-rotation-state sample of one interior-subarray port occupies about 974 kB in our implementation, and the total offline dictionary memory is about 370 MB when five representative subarray classes are used. This storage is a one-time hardware-dependent offline cost and does not scale linearly with the total number of subarrays once class sharing is exploited.

Finally, due to the inherently slow response of mechanically rotated antennas and the associated optimization complexity, the proposed RA-BS framework is not intended for slot-level mechanical re-optimization. Instead, a more practically relevant implementation is a two-timescale architecture, in which the BS/subarray orientations are updated on a slow timescale based on user distribution [12] or large-scale/statistical CSI [15], while fast-timescale adaptation is handled by conventional digital precoding/combining once the array orientations are fixed. This operating mode substantially relaxes both the online optimization burden and the real-time response requirement of the mechanical system. In this sense, the runtimes reported in Table V should be interpreted as online algorithmic cost indicators, rather than hardware-optimized per-slot deployment latencies.

VI. CONCLUSION

This paper developed a comprehensive modeling and optimization framework for polarization- and mutual-coupling-aware RA wireless communication systems. Moving beyond simplified scalar-gain models, we introduced a rotation-explicit CVEP formulation that captures the phase and polarization variations inherent in mechanically rotated arrays. By embedding these CVEPs into the 3GPP 38.901 spatial channel model and calibrating mutual coupling via orientation-dependent ERPs, the proposed method enables accurate, physically consistent evaluation of RA behavior under arbitrary rotations. On this basis, a compact dual-polarized RA antenna and RA-BS prototype were designed and validated, demonstrating stable radiation, miniaturization and high FBR. Simulations and measurements validate that the proposed EM model accurately captures rotation-induced variation in gain, polarization, and pattern shape. On this foundation, we formulated a constrained sum-rate maximization problem and proposed a feasible-direction method to jointly optimize BS and subarray orientations under mechanical limits. Our simulations across 3GPP UMi NLoS scenario confirmed that the proposed RA-BS architecture significantly outperforms the fixed BS framework and conventional RA models that neglect phase, polarization, or mutual coupling. These findings underscore the necessity of incorporating antenna radiation phase, polarization, and mutual coupling into the design and evaluation of the RA enabled wireless communication systems.

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Jianchuan Wei received the B.S. degree in detection guidance and control technology from Northwestern Polytechnical University, Xi'an, China, in 2022. He is currently pursuing the Ph.D. degree in information and communication engineering with Xi'an Jiaotong University, Xi'an. His research interests include over-the-air testing technology, programmable MIMO modeling and optimization, and near-field measurement algorithms.



Simin Song received the B.S. degree in information engineering from Xi'an Jiaotong University, Xi'an, China, in 2021, where she is currently pursuing the Ph.D. degree in electronics science and technology. Her research interests include array antennas and base station antenna design.



Zhenyu Kang received the B.Eng. degree in communication engineering from Harbin Institute of Technology in 2018, the M.Sc. degree in electrical engineering from the National University of Singapore in 2019, and the Ph.D. degree from the Department of Electrical and Computer Engineering, National University of Singapore. He is currently with Huawei Technologies Company Ltd. His research interests include MIMO communications, intelligent reflecting surfaces, UAV communications, near-field communications, electromagnetic information theory, and wireless channel modeling.



Tengjiao Wang (Member, IEEE) received the B.S. and Ph.D. degrees from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2016 and 2021, respectively. He was a Visiting Scholar with Massachusetts Institute of Technology, Cambridge, MA, USA, in 2019. He is a Principal Research Engineer with the Wireless Network RAN Research Department, Huawei Technologies Company Ltd, Shanghai, China. He has published more than 20 technical articles and holds a number of patents. His research interests

lie in the fields of 5G-Advanced wireless communications, electromagnetic information theory, visible light communications, and machine learning for wireless communications. He was a recipient of the Future Star Medal of Huawei Technologies in 2022, the Wang Daheng Optical Award in 2021, the Academic Star of Tsinghua University in 2021, the Excellent Doctoral Dissertation of Tsinghua University in 2021, and the National Scholarship in 2018. He has served as the Workshop Organizer, the Co-Chair, and the TPC Member for many international conferences, including IEEE International Conference on Communications (ICC), IEEE Global Communications Conference (GLOBECOM), IEEE Vehicular Technology Conference (VTC), and IEEE/CIC International Conference on Communications in China (ICCC).



Huaqiang Gao (Member, IEEE) received the B.E. degree in electronic and information engineering from Harbin University of Science and Technology, Harbin, China, in 2016, the B.A. degree in business English from Heilongjiang University, Harbin, in 2016, and the Ph.D. degree in electronic science and technology from Beijing University of Posts and Telecommunications, Beijing, China, in 2022. Since 2019, he has been a Visiting Ph.D. Student and a Post-Doctoral Fellow with the Antennas, Propagation and Millimeter-Wave Systems Section, Aalborg University, Aalborg, Denmark. He is currently an Assistant Professor with the School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an, China. His research interests include over-the-air (OTA) testing of wireless devices, antenna measurement, and phased array calibration.



Shanpu Shen (Senior Member, IEEE) received the bachelor's degree in communication engineering from Nanjing University of Science and Technology, Nanjing, China, in 2013, and the Ph.D. degree in electronic and computer engineering from The Hong Kong University of Science and Technology (HKUST), Hong Kong, in 2017. He was a Visiting Ph.D. Student with Microsystems Technology Laboratories, Massachusetts Institute of Technology, Cambridge, MA, USA, in 2016. He was a Post-Doctoral Fellow with HKUST from 2017 to 2018 and a Post-Doctoral Research Associate with the Imperial College London, U.K., from 2018 to 2020. He was a Research Assistant Professor with HKUST from 2020 to 2023 and a Lecturer (an Assistant Professor) with the University of Liverpool, U.K., from 2023 to 2025. He is currently an Associate Professor with the State Key Laboratory of Internet of Things for Smart City and the Department of Electrical and Computer Engineering, University of Macau, Macau. His current research interests include the Internet-of-Things, pixel antenna, reconfigurable intelligent surface, electromagnetic information theory, RF energy harvesting, and wireless power transfer. He was selected as the 2024 IEEE Antenna and Propagation Society Young Professional Ambassador. He is serving as an Associate Editor for IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION.



Xiaoming Chen (Senior Member, IEEE) received the B.Sc. degree in electrical engineering from Northwestern Polytechnical University, Xi'an, China, in 2006, and the M.Sc. and Ph.D. degrees in electrical engineering from the Chalmers University of Technology, Gothenburg, Sweden, in 2007 and 2012, respectively. From 2013 to 2014, he was a Post-Doctoral Researcher with the Chalmers University of Technology. From 2014 to 2017, he was with Qamcom Research and Technology AB, Gothenburg, Sweden. Since 2017, he has been a Professor with Xi'an Jiaotong University, Xi'an. His research areas include MIMO antennas, over-the-air testing, reverberation chambers, and electromagnetic information theory. He received the Outstanding AE/TE Awards eight times from 2018 to 2025 and the International Union of Radio Science (URSI) Young Scientist Awards in 2017 and 2018. He serves as an Associate Editor (AE) for IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION and a Track Editor for IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS.